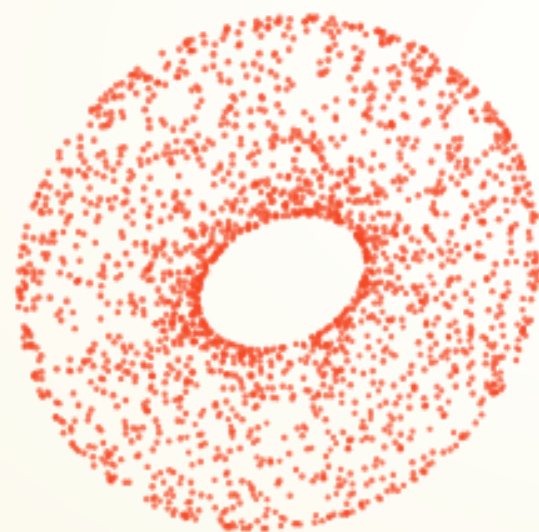


PERSISTENT π_1

& LOWER-CENTRAL-SERIES BARCODES

GONZALO ORTEGA CARPINTERO

gonzalo.ortega-carpintero@city.ac.uk

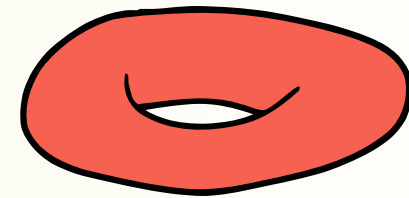


A joint work with Ximena Fernández at

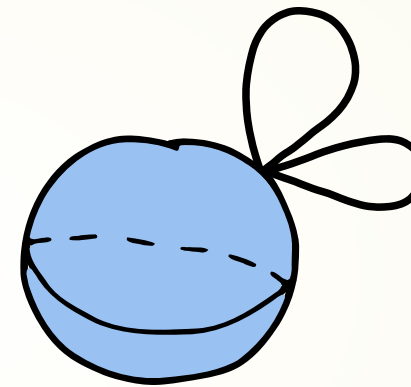
CITY
ST GEORGE'S
UNIVERSITY OF LONDON

YTM 2026 - Monday, 29th June

THE LIMITATIONS OF H_*



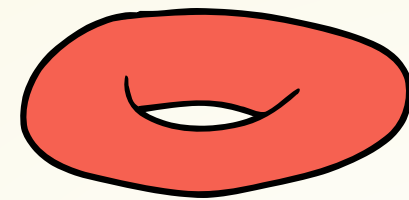
$$T = S_1 \times S_1$$



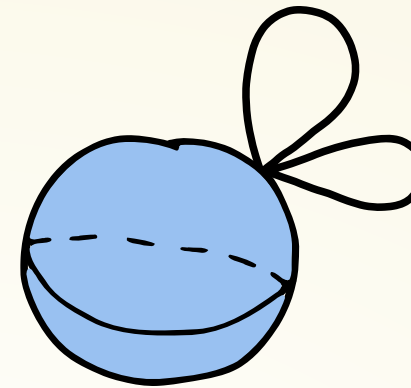
$$X = S^2 \vee S^1 \vee S^1$$

All computations will be made with coefficients in a field $\mathbb{k}$.

THE LIMITATIONS OF H_*



$$T = S_1 \times S_1$$



$$X = S^2 \vee S^1 \vee S^1$$

$$H_0(T) = \mathbb{k} = H_0(X)$$

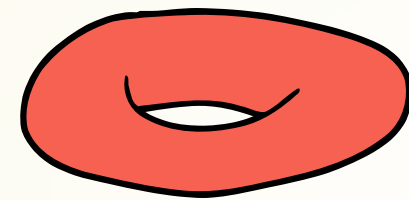
$$H_1(T, \mathbb{k}) = \mathbb{k} \oplus \mathbb{k} = H_1(X, \mathbb{k})$$

$$H_2(T) = \mathbb{k} = H_2(X)$$

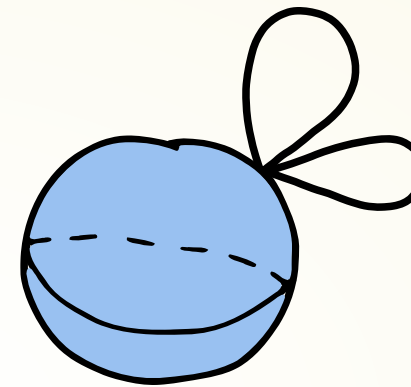
$$H_n(T) = 0 = H_n(X),$$

$$n \geq 3$$

THE LIMITATIONS OF H_*



$$T = S_1 \times S_1$$

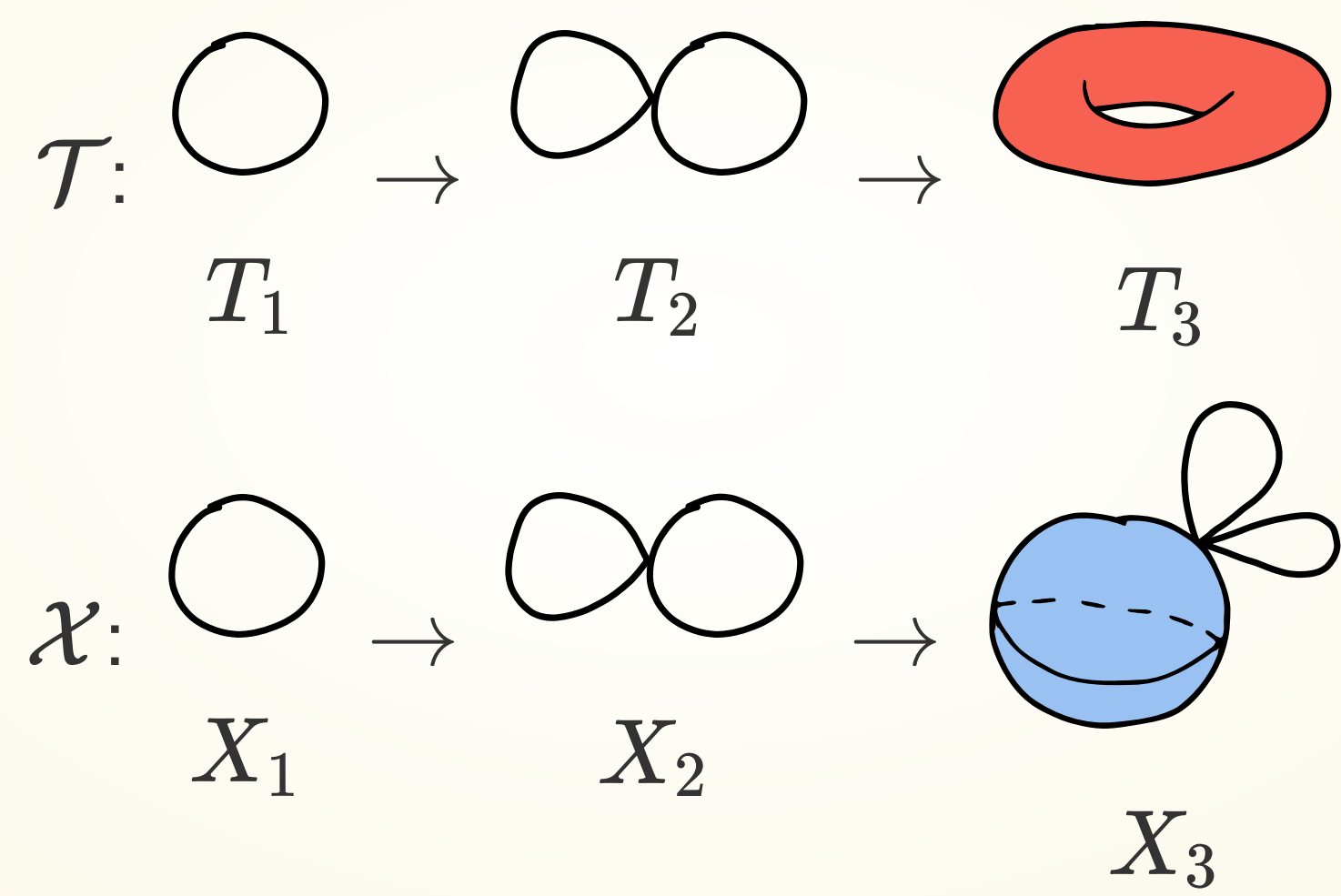


$$X = S^2 \vee S^1 \vee S^1$$

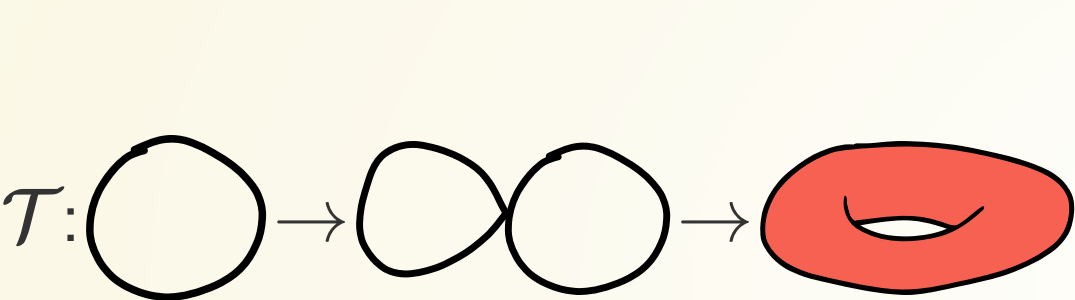
$$H_1(T, \mathbb{k}) = \mathbb{k} \oplus \mathbb{k} = H_1(X, \mathbb{k})$$

$$\pi_1(T) = \mathbb{Z} \oplus \mathbb{Z} \neq \mathbb{Z} * \mathbb{Z} = \pi_1(X)$$

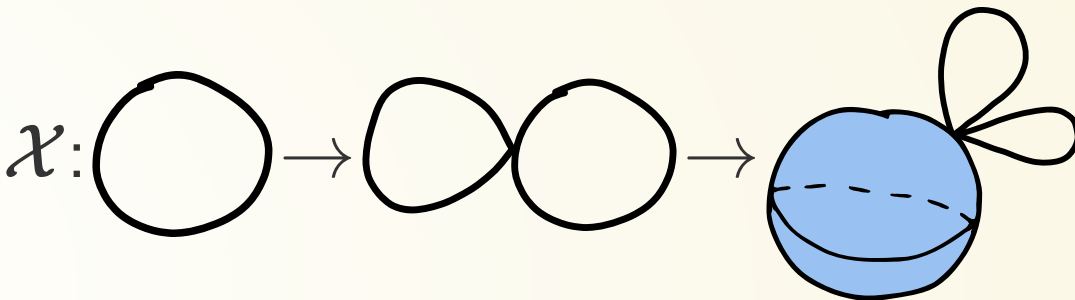
THE LIMITATIONS OF PERSISTENT H_*



THE LIMITATIONS OF PERSISTENT H_*

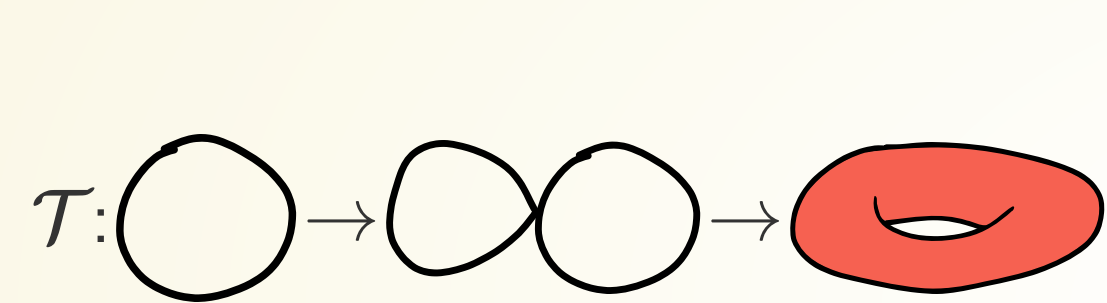


T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3
$\downarrow$		$\downarrow$		$\downarrow$
$H_1(T_1, \mathbb{k})$	$\longrightarrow$	$H_1(T_2, \mathbb{k})$	$\longrightarrow$	$H_1(T_3, \mathbb{k})$

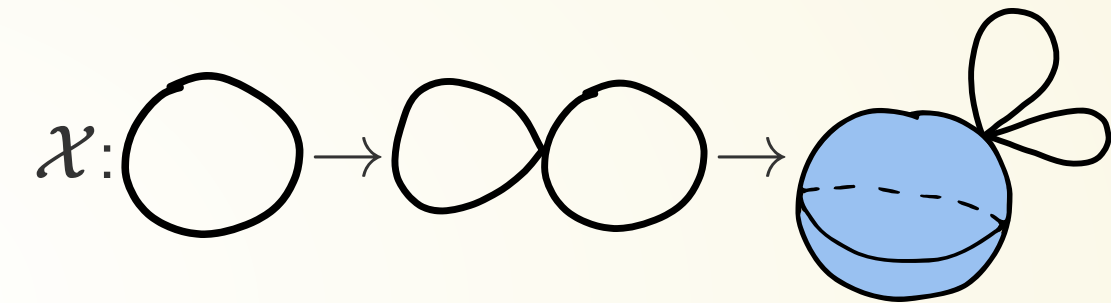


X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow$		$\downarrow$		$\downarrow$
$H_1(X_1, \mathbb{k})$	$\longrightarrow$	$H_1(X_2, \mathbb{k})$	$\longrightarrow$	$H_1(X_3, \mathbb{k})$

THE LIMITATIONS OF PERSISTENT H_*

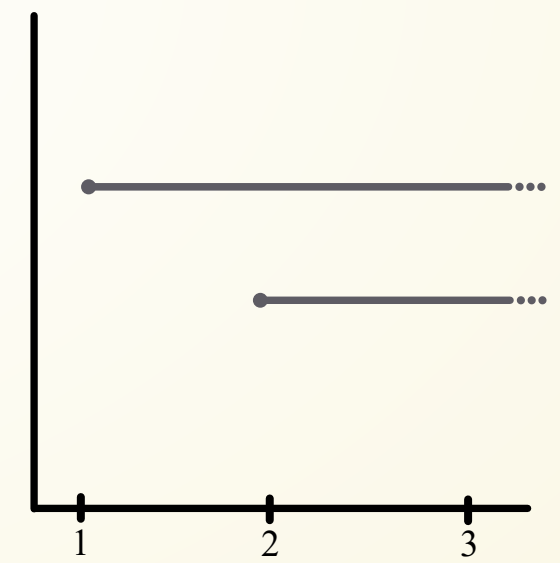
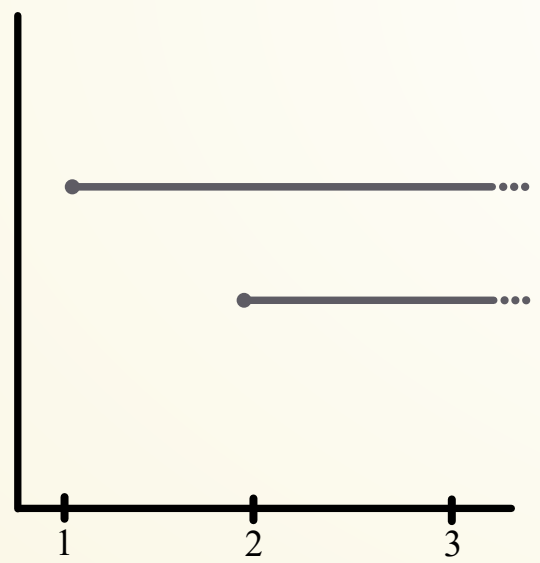
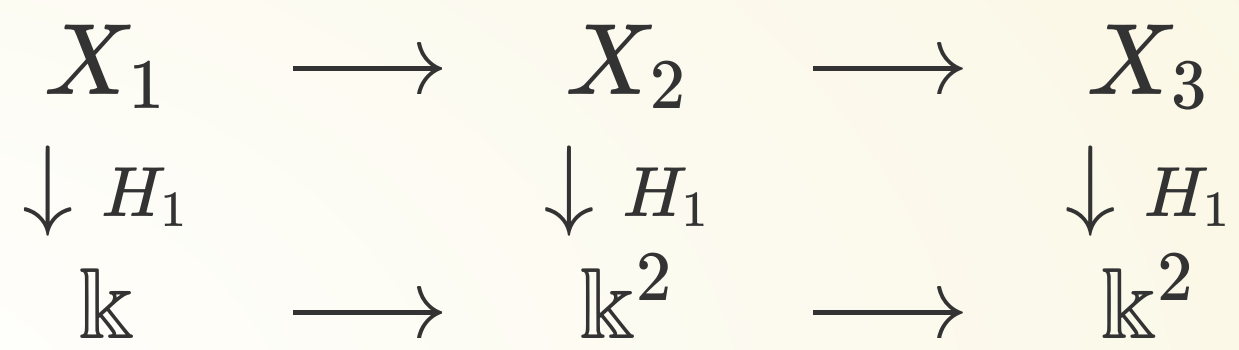
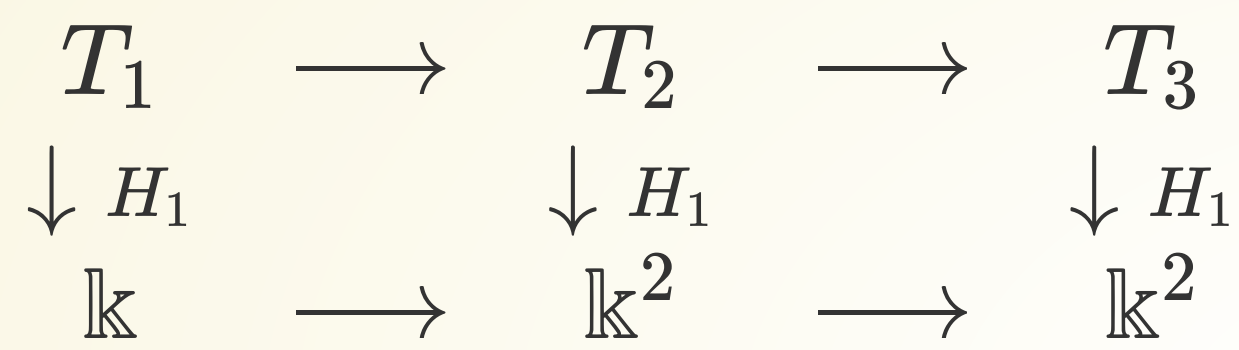
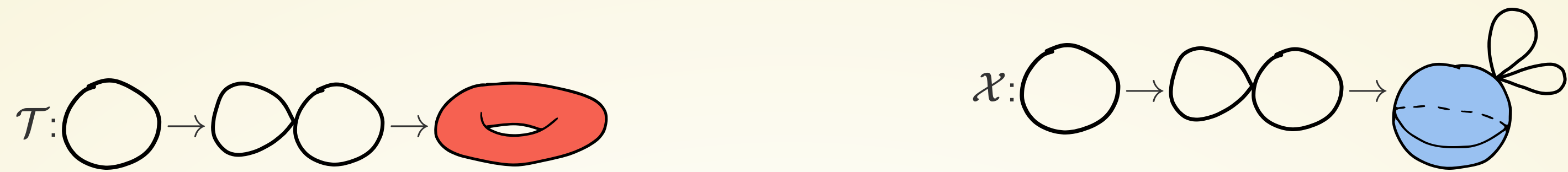


T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3
$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$
$\mathbb{k}$	$\longrightarrow$	$\mathbb{k}^2$	$\longrightarrow$	$\mathbb{k}^2$



X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$
$\mathbb{k}$	$\longrightarrow$	$\mathbb{k}^2$	$\longrightarrow$	$\mathbb{k}^2$

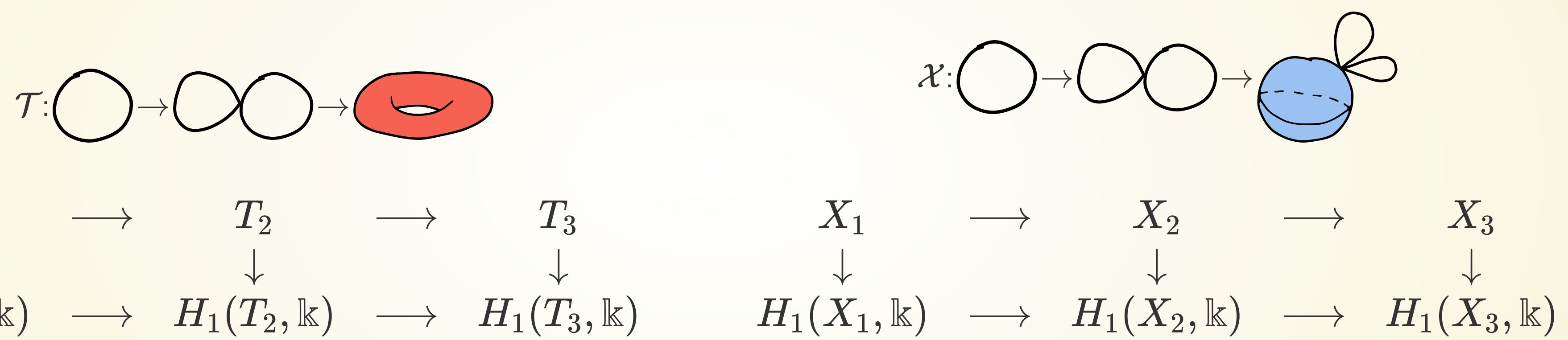
THE LIMITATIONS OF PERSISTENT H_*



$PH_1(\mathcal{T}) = PH_1(\mathcal{X})$

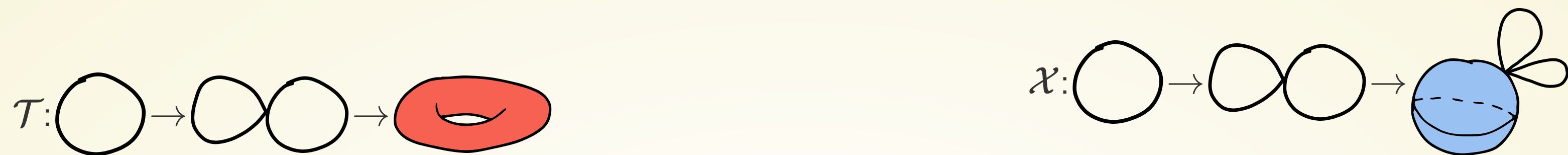
All barcodes coincide.

THE LIMITATIONS OF PERSISTENT H_*



WHAT WOULD WE LIKE TO DO?

THE LIMITATIONS OF PERSISTENT H_*



T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3		X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$
$H_1(T_1, \mathbb{k})$	$\longrightarrow$	$H_1(T_2, \mathbb{k})$	$\longrightarrow$	$H_1(T_3, \mathbb{k})$		$H_1(X_1, \mathbb{k})$	$\longrightarrow$	$H_1(X_2, \mathbb{k})$	$\longrightarrow$	$H_1(X_3, \mathbb{k})$

PERSISTENT π_1

T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3		X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$
$\pi_1(T_1)$	$\longrightarrow$	$\pi_1(T_2)$	$\longrightarrow$	$\pi_1(T_3)$		$\pi_1(X_1)$	$\longrightarrow$	$\pi_1(X_2)$	$\longrightarrow$	$\pi_1(X_3)$

Convention: All spaces are path-connected and contain a common basepoint.

CAN π_1 BE ALGORITHMICALLY COMPUTED?

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

*We can compute a **presentation** of π_1 of a CW complex.

*(All CW complexes will be **finite** and **path-connected** in this presentation.)*

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

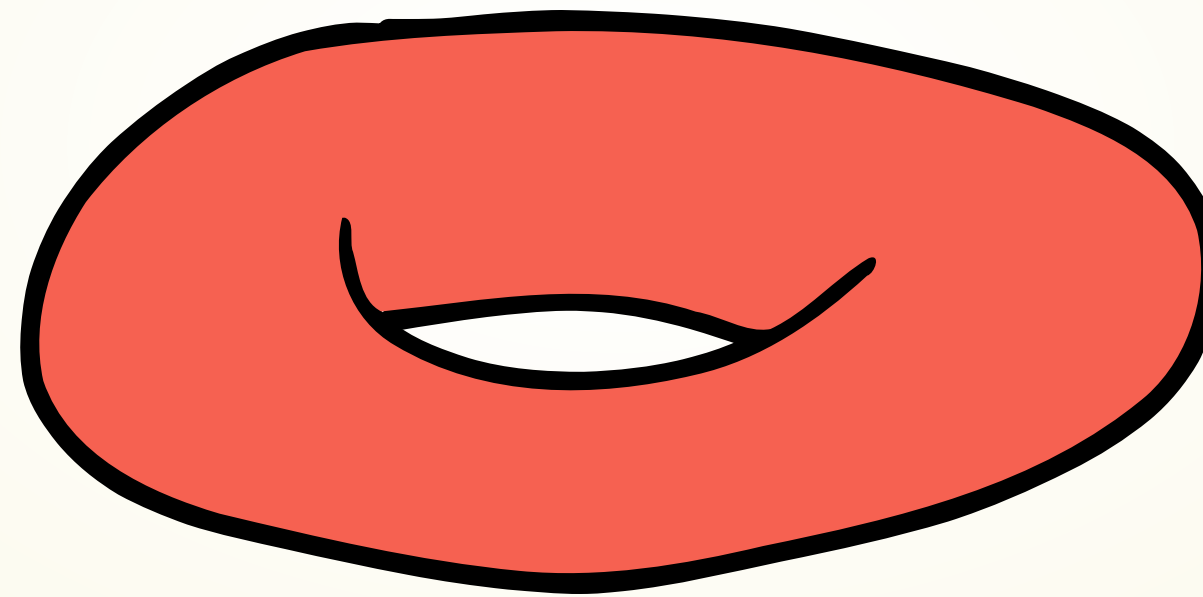
*We can compute a **presentation** of π_1 of a CW complex.

Naive computation: Using the **spanning tree** algorithm.

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

*We can compute a **presentation** of π_1 of a CW complex.

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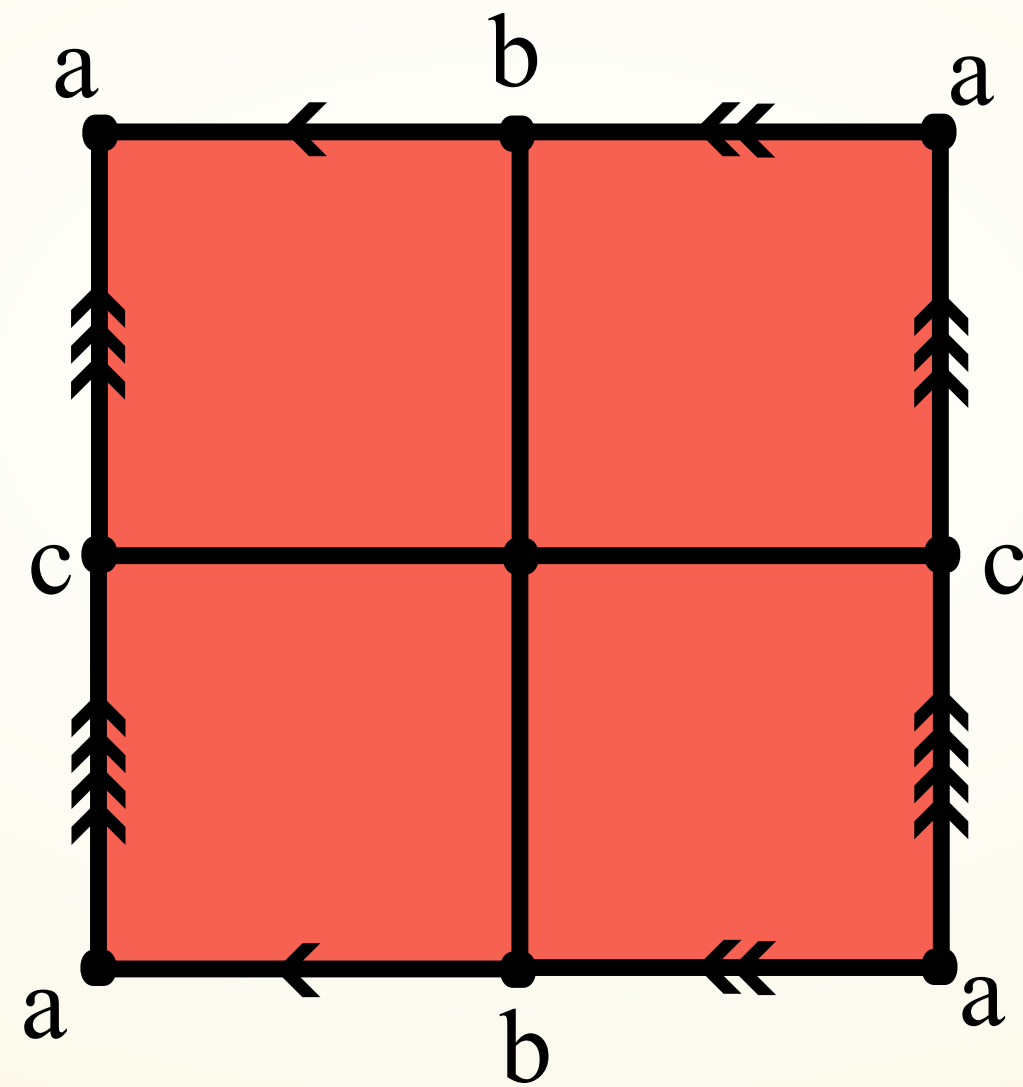


T

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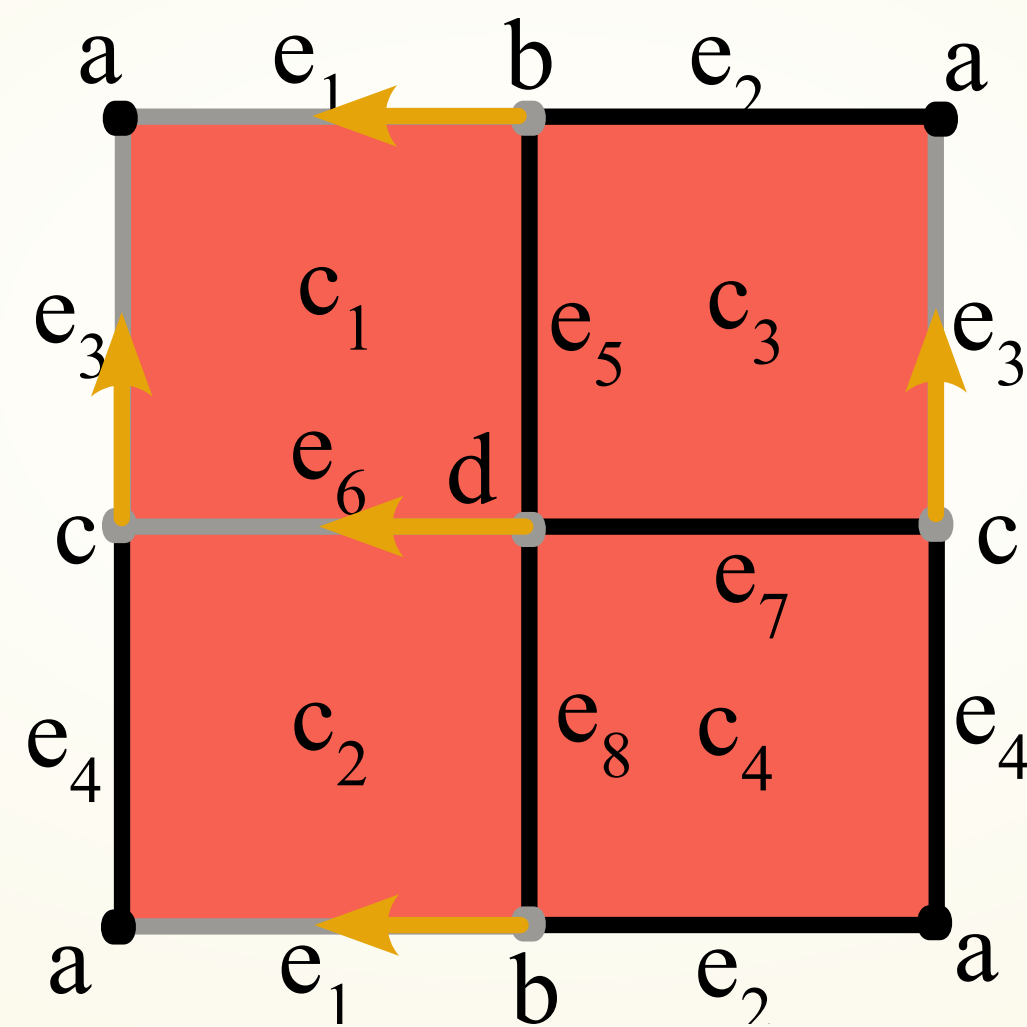


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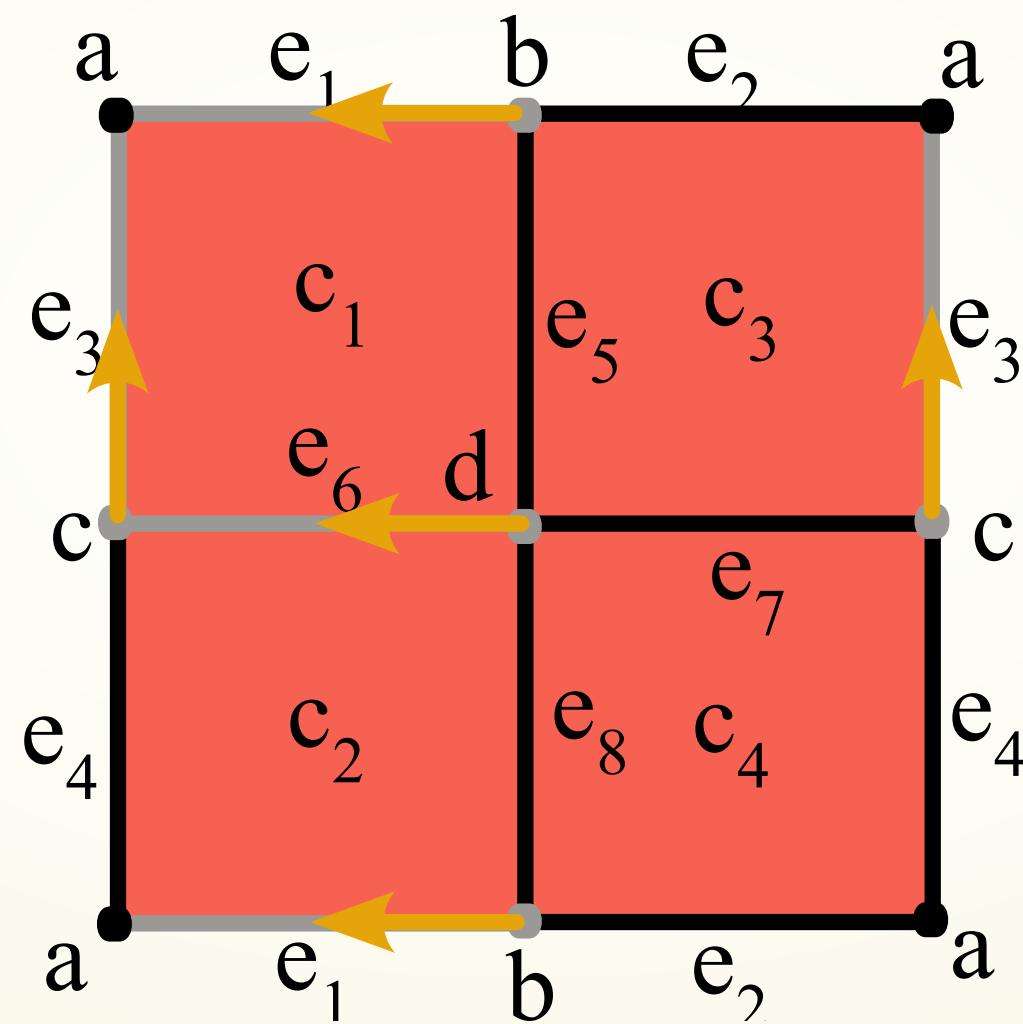


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CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

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Naive computation: Using the **spanning tree** algorithm.



$$T \cong \langle e_2, e_4, e_5, e_7, e_8 \mid e_5, e_2^{-1}e_7, e_5^{-1}, e_7^{-1}e_4^{-1}e_2e_8, e_8^{-1}e_4 \rangle$$

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

*We can compute a **presentation** of π_1 of a CW complex.

Naive computation: Using the **spanning tree** algorithm.

Efficient computation: Using **discrete Morse theory** to obtain a reduced presentation.

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

*We can compute a **presentation** of π_1 of a CW complex.

Efficient computation: Using **discrete Morse theory** to obtain a reduced presentation.

Let K be a regular CW complex and let M be an **acyclic matching** on the cells of K .

Theorem (Forman, 1995; Chari 2000): K is homotopy equivalent to a CW complex with exactly one cell of dimension p for **each critical** (i.e., unmatched) cell of dimension p .

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

*We can compute a **presentation** of π_1 of a CW complex.

Efficient computation: Using **discrete Morse theory** to obtain a reduced presentation.

Let K be a regular CW complex and let M be an acyclic matching
(with exactly one critical 0-cell).

Theorem (Fernández, 2024): A presentation $\mathcal{P} = \langle X \mid R \rangle$ of $\pi_1(K)$
with

$$\#X = \#1\text{-dim critical cells}$$

$$\#R = \#2\text{-dim critical cells}$$

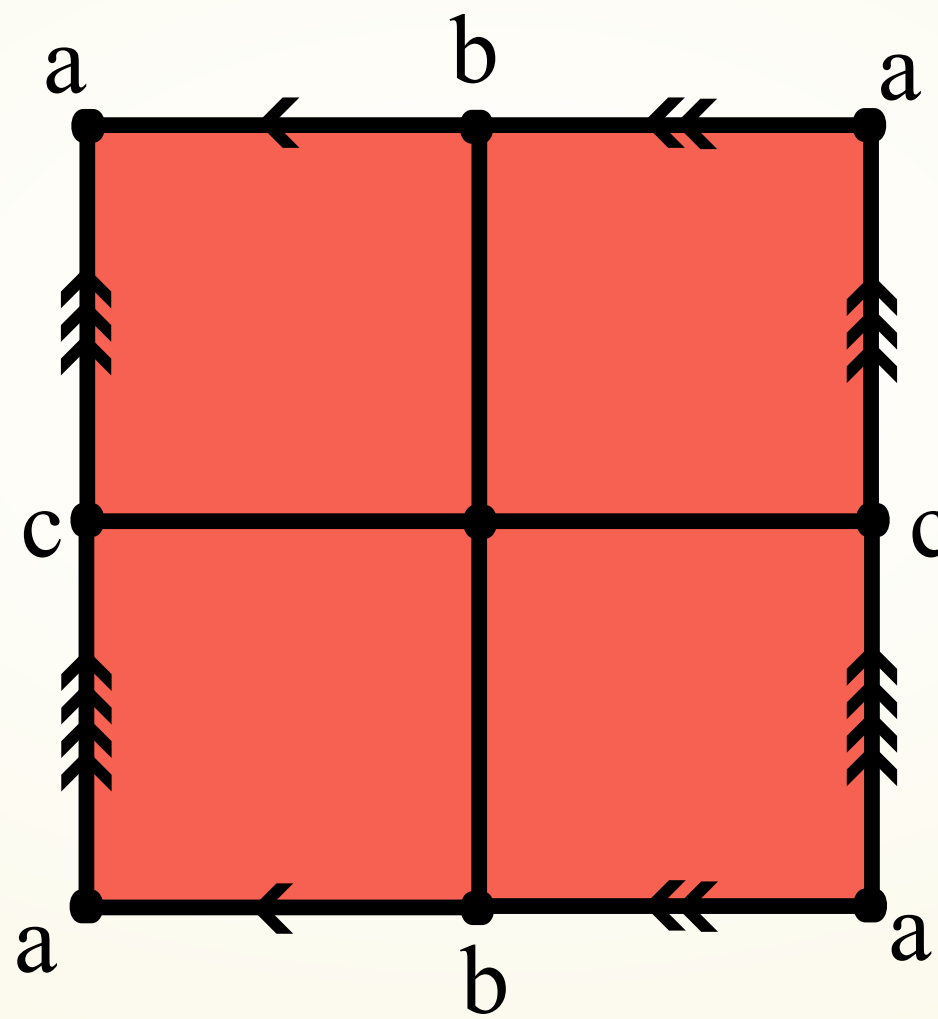
can be computed with complexity $\mathcal{O}(N^2)$, where

$$N = \#\text{cells in the 2-skeleton of } K.$$

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Efficient computation: Using **discrete Morse theory** to obtain a reduced presentation.

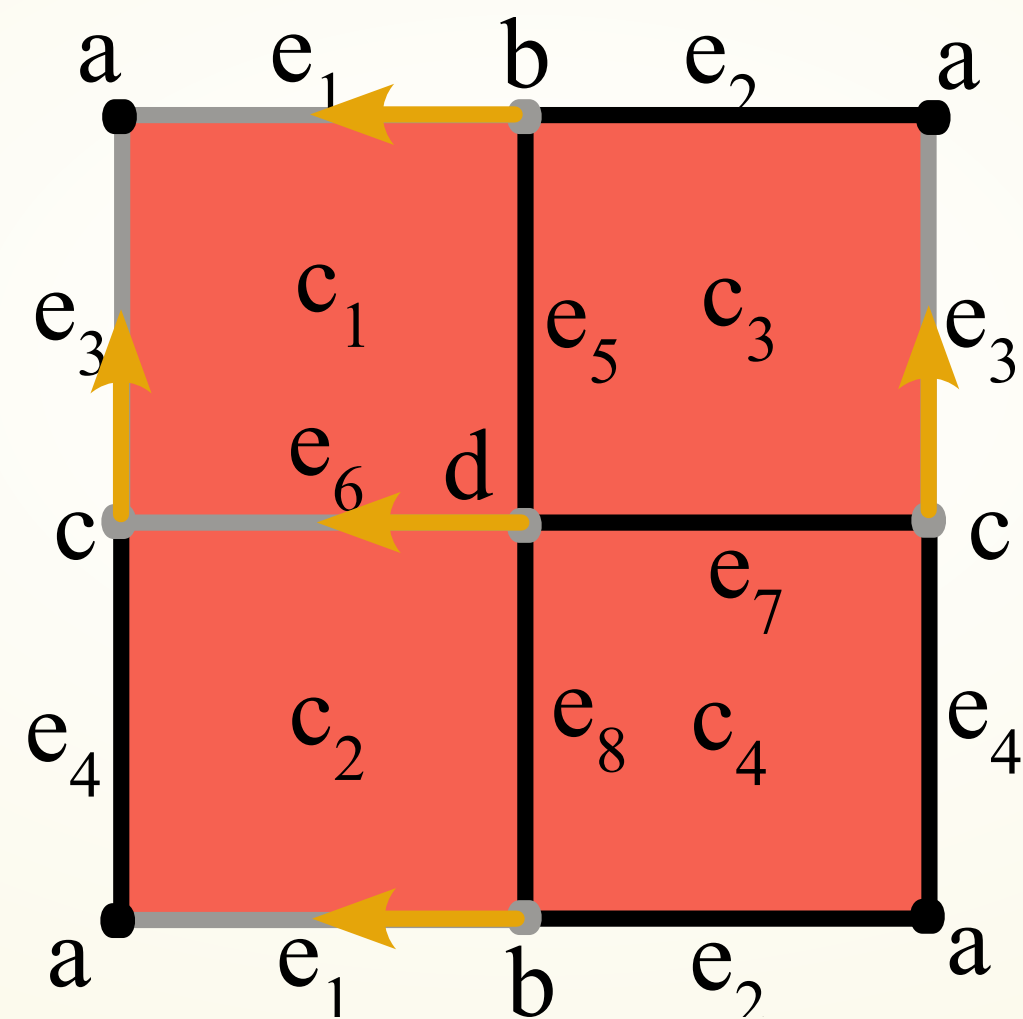


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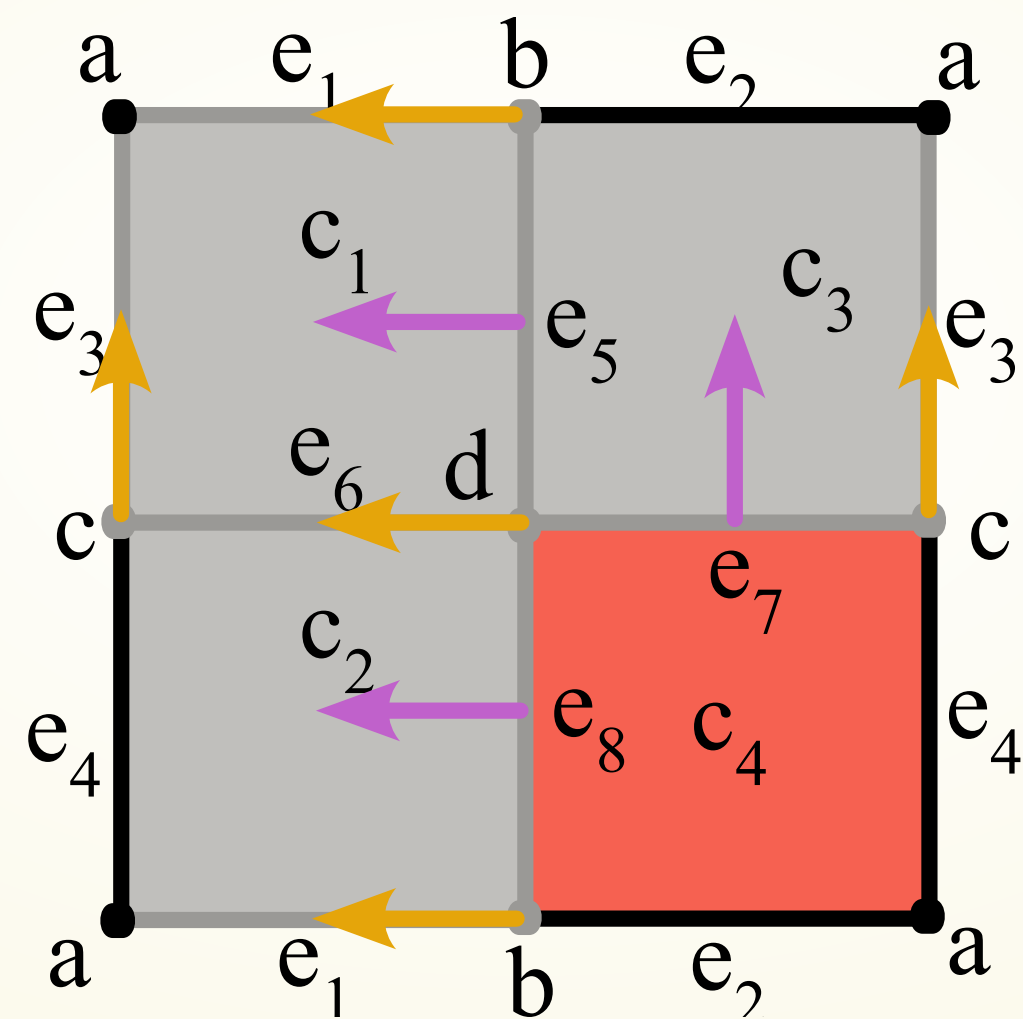


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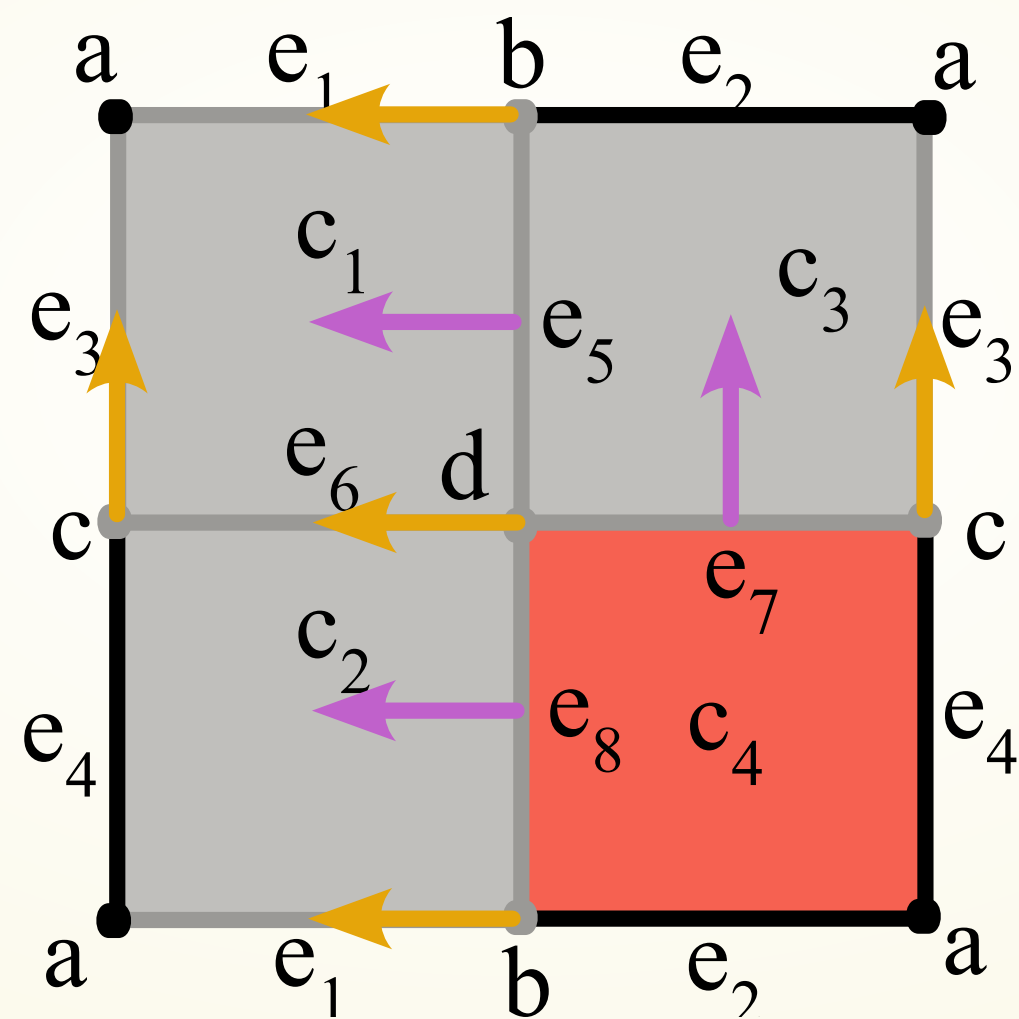


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$$T \cong \langle e_2, e_4 \mid e_2^{-1} e_4^{-1} e_2 e_4 \rangle$$

CAN π_1 BE ALGORITHMICALLY COMPUTED? **YES***

CAN π_1 BE COMPUTED OVER A FILTRATION?

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED?

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

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CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED? YES*

Let $K_\bullet$ be a path-connected based filtration of finite CW-complexes.

$$\begin{array}{ccccccc} K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots & \longrightarrow & K_n \\ \downarrow & & \downarrow & & & & \downarrow \\ \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots & \longrightarrow & \pi_1(K_n) \end{array}$$

Naive computation: Compute π_1 presentations **independently** over every element in the filtration.

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED? YES*

Let $K_\bullet$ be a path-connected based filtration of finite CW-complexes.

$$\begin{array}{ccccccc} K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots & \longrightarrow & K_n \\ \downarrow & & \downarrow & & & & \downarrow \\ \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots & \longrightarrow & \pi_1(K_n) \end{array}$$

Naive computation: Compute π_1 presentations **independently** over every element in the filtration.

Efficient computation: Perform a **single computation** over the whole filtration.

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED? YES*

Efficient computation: Perform a **single computation** over the whole filtration.

Theorem (Fernández, O., 2026): Let $K_\bullet$ be a based filtration of finite regular CW complexes and let $M_\bullet$ be a filtration compatible acyclic matching with exactly one critical 0-cell at each level. We can compute presentations

$$\mathcal{P}_i = \langle X_i \mid R_i \rangle \cong \pi_1(K_i) \text{ with}$$

$$\begin{aligned} \#X_i &= \#1\text{-dim critical cells of } K_i, \\ \#R_i &= \#2\text{-dim critical cells of } K_i \end{aligned}$$

and compatible morphisms $\psi_{i \rightarrow j} : \mathcal{P}_i \rightarrow \mathcal{P}_j$ such that

$$\begin{array}{ccc} K_i & \hookrightarrow & K_j \\ \downarrow & & \downarrow \\ \mathcal{P}_i & \xrightarrow{\psi_{i \rightarrow j}} & \mathcal{P}_j \end{array}$$

commutes, in time $\mathcal{O}(N^2)$, $N = \#$ cells in the 2-skeleton of K_n .

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED? YES*

Efficient computation: Perform a **single computation** over the whole filtration.

SAGE Math implementation

```
def filtrated_fundamental_group(self, fil, verbose = True):
    S = set()
    R = []
    filtrated_fundamental_group = []
    maps = [[None] * (fil+1) for _ in range(len(self.cells_lists[1]))]

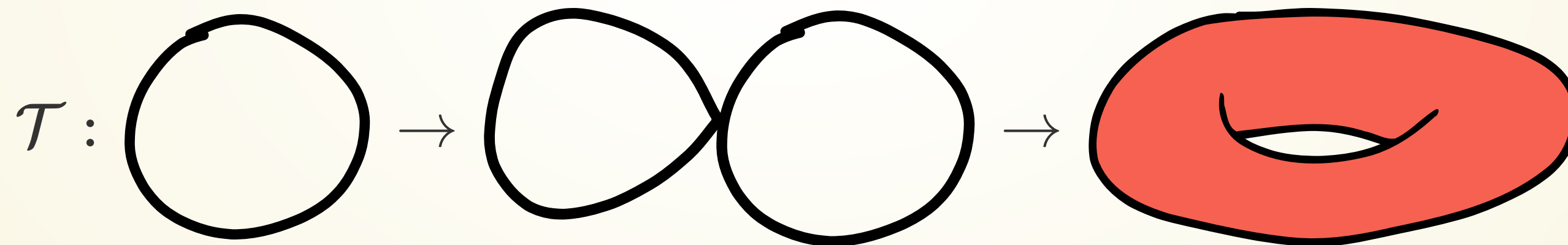
    for f in range(0, fil+1):

        for cell in self.filtration_unmatched_2_cells[f]:
            R.append(cell.boundary_cells)

        for cell in self.filtration_matched_1_cells_with_0_cells[f]:
            cell.equivalence = '1'
            maps[cell_count][f] = 1
```

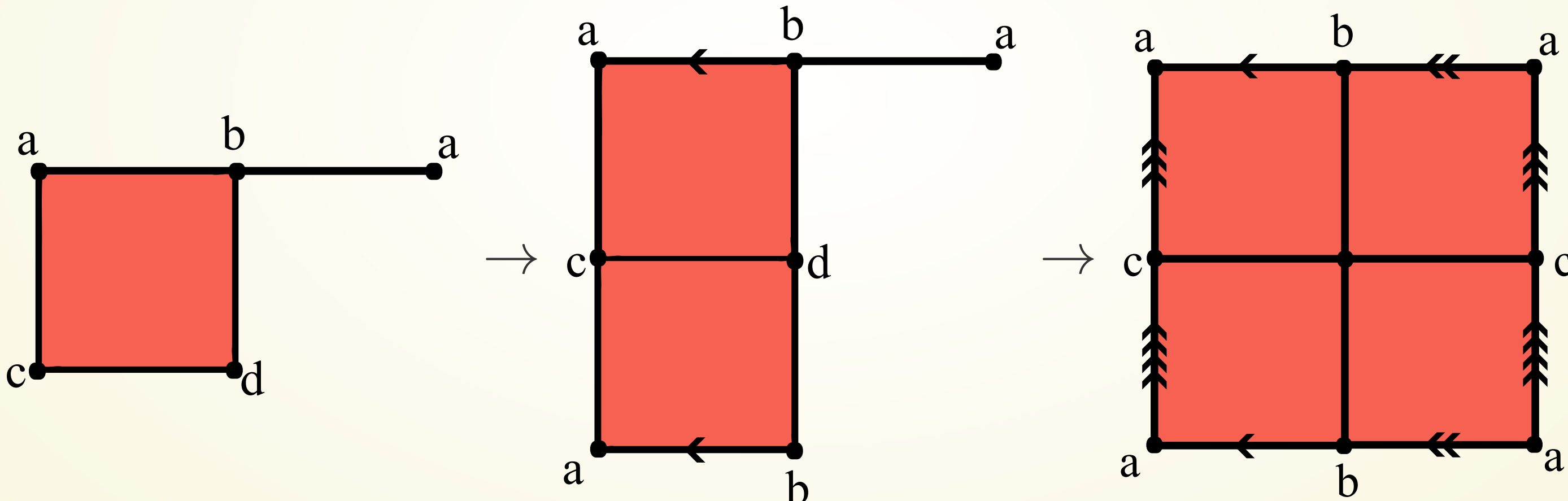
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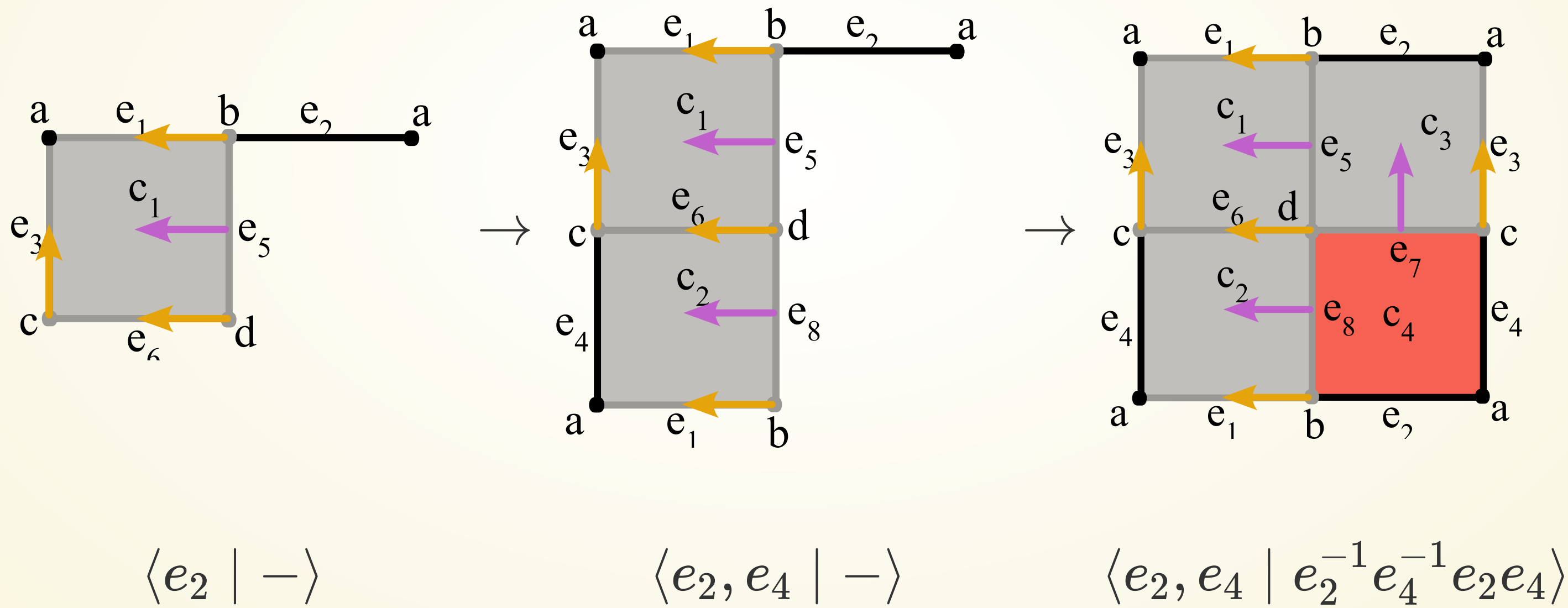
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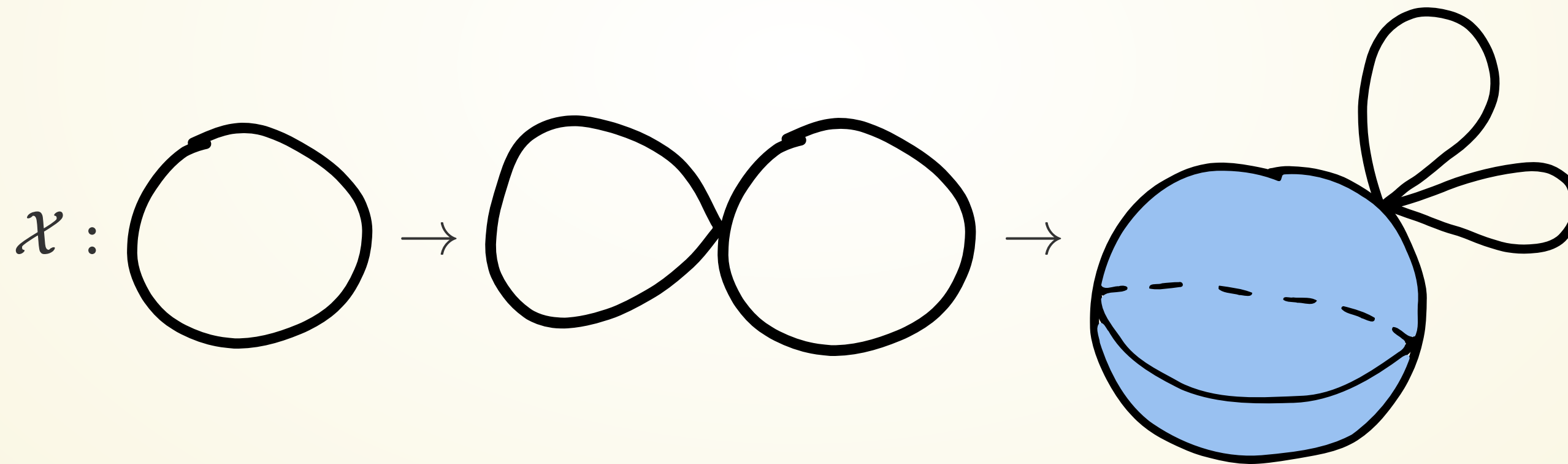
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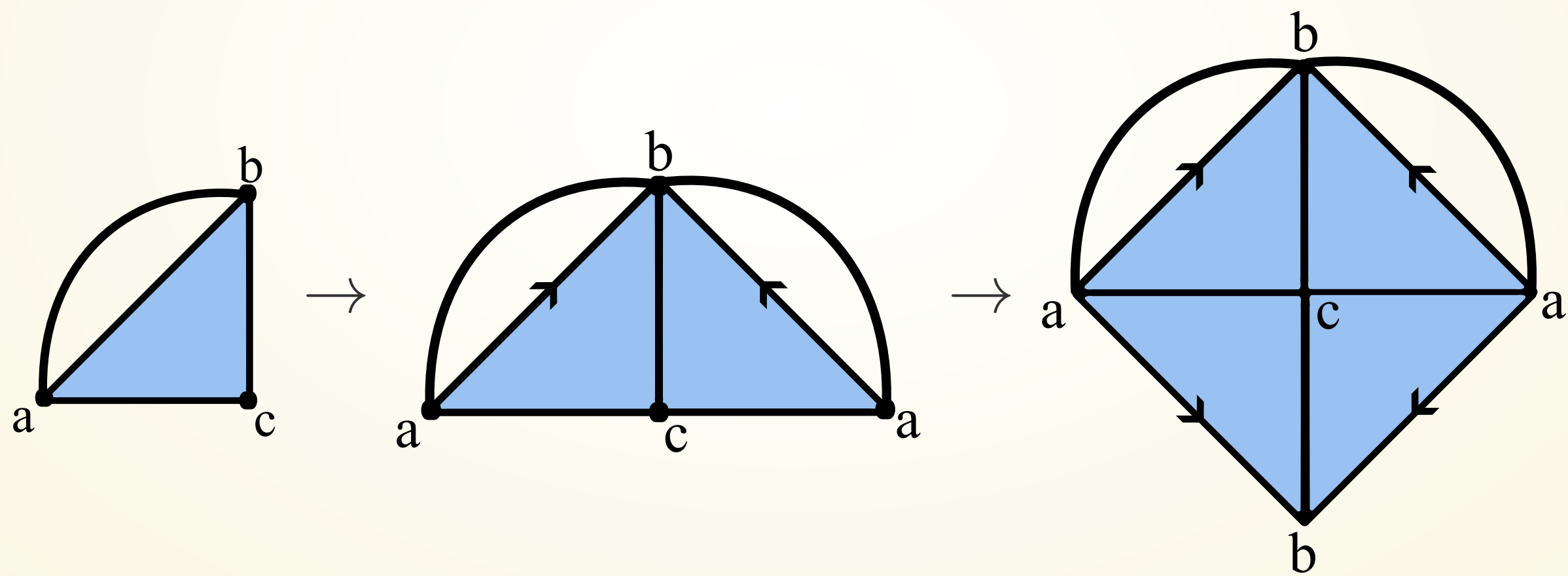
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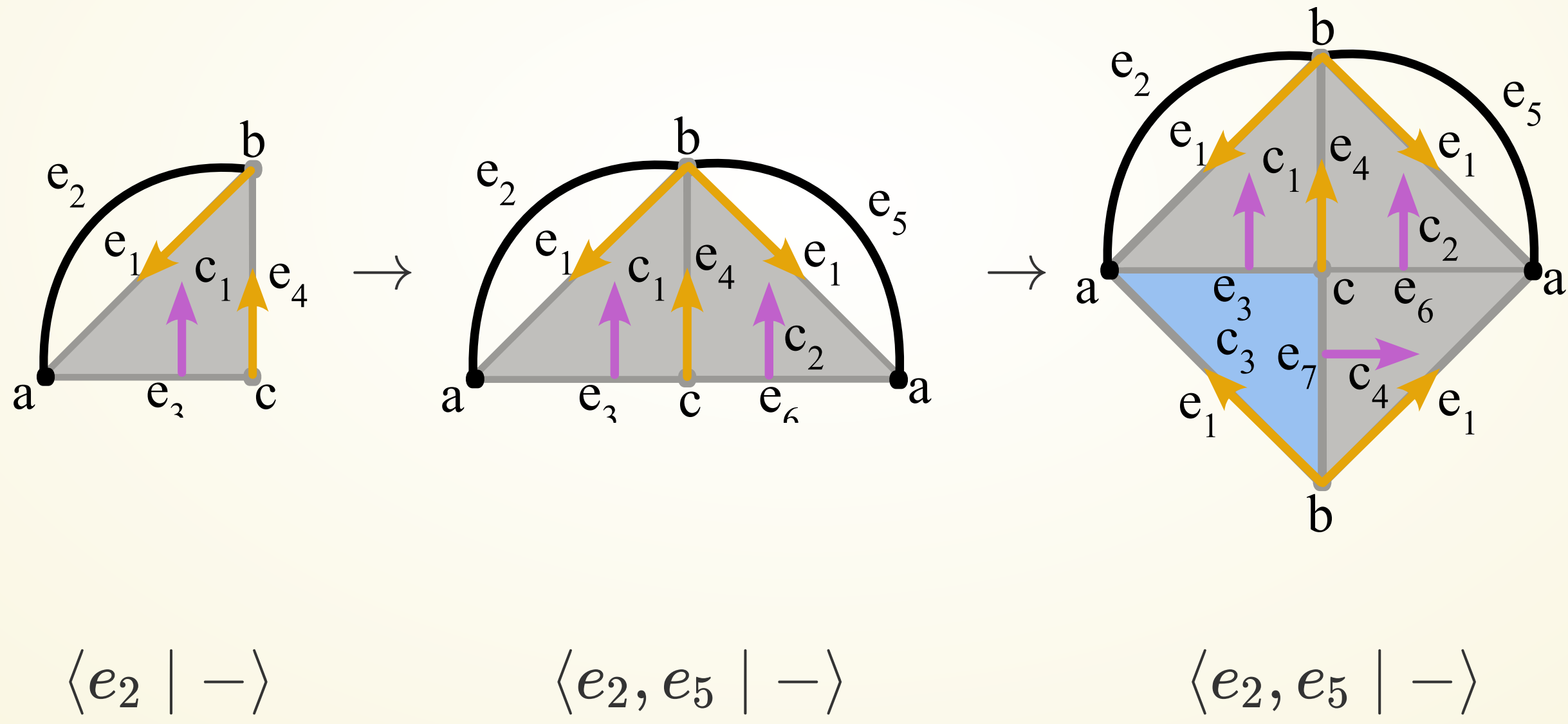
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CAN PERSISTENT π_1 BE COMPUTED? **YES***



CAN π_1 BE ALGORITHMICALLY COMPUTED? **YES***

CAN PERSISTENT π_1 BE COMPUTED? **YES***

$$P\pi_1(\mathcal{T}) = \langle t_1 \mid - \rangle \rightarrow \langle t_1, t_2 \mid - \rangle \rightarrow \langle t_1, t_2 \mid t_1^{-1}t_2^{-1}t_1t_2 \rangle,$$

$$P\pi_1(\mathcal{X}) = \langle x_1 \mid - \rangle \rightarrow \langle x_1, x_2 \mid - \rangle \rightarrow \langle x_1, x_2 \mid - \rangle.$$

CAN π_1 BE ALGORITHMICALLY COMPUTED? YES*

CAN PERSISTENT π_1 BE COMPUTED? YES*

$$P\pi_1(\mathcal{T}) = \langle t_1 \mid - \rangle \rightarrow \langle t_1, t_2 \mid - \rangle \rightarrow \langle t_1, t_2 \mid t_1^{-1}t_2^{-1}t_1t_2 \rangle,$$

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CAN WE MAKE π_1 BARCODES AS IN H_* ?

CAN π_1 BE ALGORITHMICALLY COMPUTED? **YES***

CAN PERSISTENT π_1 BE COMPUTED? **YES***

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CAN WE MAKE π_1 BARCODES AS IN H_* ? **NO***

- The groups $\pi_1(K_\alpha)$ in $K_\bullet \rightarrow P\pi_1(K)_\bullet$.

$$\pi_1(K_1) \longrightarrow \pi_1(K_2) \longrightarrow \cdots \longrightarrow \pi_1(K_n)$$

are generally **non-abelian**.

CAN π_1 BE ALGORITHMICALLY COMPUTED? **YES***

CAN PERSISTENT π_1 BE COMPUTED? **YES***

$$P\pi_1(\mathcal{T}) = \langle t_1 \mid - \rangle \rightarrow \langle t_1, t_2 \mid - \rangle \rightarrow \langle t_1, t_2 \mid t_1^{-1}t_2^{-1}t_1t_2 \rangle,$$

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$$\pi_1(K_1) \longrightarrow \pi_1(K_2) \longrightarrow \cdots \longrightarrow \pi_1(K_n)$$

are generally **non-abelian**.

- Finite presentations are **not group invariants** in general.
- Ordinary barcode decompositions do not apply.

IDEA:

LOOK FOR AN ABELIAN INVARIANT OF π_1

LOWER CENTRAL QUOTIENTS

LOWER CENTRAL QUOTIENTS

Definition: The **lower central series** of a group G is the descending series of normal subgroups

$$G = \gamma_1(G) \supseteq \gamma_2(G) \supseteq \cdots \supseteq \gamma_n(G) \supseteq \cdots$$

where

$$\gamma_{n+1}(G) = [\gamma_n(G), G] = \langle [h, g] : h \in \gamma_n(G), g \in G \rangle.$$

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where

$$\gamma_{n+1}(G) = [\gamma_n(G), G] = \langle [h, g] : h \in \gamma_n(G), g \in G \rangle.$$

Definition: We define the n^{th} -**lower central quotient** of G as

$$lcq_n(G) := \frac{\gamma_n(G)}{\gamma_{n+1}(G)}.$$

LOWER CENTRAL QUOTIENTS

Example: Let $G = \pi_1(K)$.

LOWER CENTRAL QUOTIENTS

Example: Let $G = \pi_1(K)$.

$$lcq_1(G) = \frac{G}{[G, G]} \cong H_1(K, \mathbb{k}),$$

$$lcq_2(G) = \frac{[G, G]}{[[G, G], G]},$$

$$lcq_3(G) = \frac{[[G, G], G]}{[[[G, G], G], G]},$$

...

LOWER CENTRAL QUOTIENTS

Example (Wedge of circles): Let $K = \bigvee_n S^1$, $G = \pi_1 (K) = F_n$.

$$lcq_1(F_n) \cong \mathbb{Z}^n,$$

$$lcq_2(F_n) \cong \mathbb{Z}^{\binom{n}{2}},$$

$$lcq_3(F_n) \cong \mathbb{Z}^{n\binom{n}{2} - \binom{n}{3}}.$$

LOWER CENTRAL QUOTIENTS

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$$lcq_1(F_n) \cong \mathbb{Z}^n,$$

$$lcq_2(F_n) \cong \mathbb{Z}^{\binom{n}{2}},$$

$$lcq_3(F_n) \cong \mathbb{Z}^{n\binom{n}{2} - \binom{n}{3}}.$$

lcq_n measures n -fold commutativity features.

LOWER CENTRAL QUOTIENTS

For every n , $lcq_n(G)$ is abelian.

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The complexity of the computation of $lcq_n(G)$ depends on the number of generators and relations of the presentation of G we are using.

PERSISTENT LOWER CENTRAL QUOTIENTS

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Definition: Let $\mathbb{k}$ a fixed field. We define the $\mathbb{k}$ -tensored n^{th} -lower central quotient as

$$lcq_n^{\mathbb{k}} := lcq_n \otimes_{\mathbb{Z}} \mathbb{k}$$

PERSISTENT LOWER CENTRAL QUOTIENTS

$$\begin{array}{ccccccc}
 K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots \longrightarrow & K_n \\
 \downarrow & & \downarrow & & & \downarrow \\
 \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots \longrightarrow & \pi_1(K_n) \\
 \downarrow \xi_1 & & \downarrow \xi_2 & & & \downarrow \xi_n \\
 lcq_n^{\mathbb{k}}(\pi_1(K_1)) & \longrightarrow & lcq_n^{\mathbb{k}}(\pi_1(K_2)) & \longrightarrow & \cdots \longrightarrow & lcq_n^{\mathbb{k}}(\pi_1(K_n))
 \end{array}$$

PERSISTENT LOWER CENTRAL QUOTIENTS

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 K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots \longrightarrow & & K_n \\
 \downarrow & & \downarrow & & & & \downarrow \\
 \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots \longrightarrow & & \pi_1(K_n) \\
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 lcq_n^{\mathbb{k}}(\pi_1(K_1)) & \longrightarrow & lcq_n^{\mathbb{k}}(\pi_1(K_2)) & \longrightarrow & \cdots \longrightarrow & & lcq_n^{\mathbb{k}}(\pi_1(K_n))
 \end{array}$$

Here we obtain a functor

$$\xi : \mathbf{Grp} \rightarrow \mathbf{Vect}_{\mathbb{k}}$$

$$G \mapsto \frac{\gamma_n(G)}{\gamma_{n+1}(G)} \otimes_{\mathbb{Z}} \mathbb{k}$$

PERSISTENT LOWER CENTRAL QUOTIENTS

Every $lcq_n(\pi_1(K_\alpha))$ is abelian so

$$lcq_n^{\mathbb{k}}(\pi_1(K_1)) \longrightarrow lcq_n^{\mathbb{k}}(\pi_1(K_2)) \longrightarrow \cdots \longrightarrow lcq_n^{\mathbb{k}}(\pi_1(K_n))$$

is a **persistent module** over the field $\mathbb{k}$ (ie. a graded $\mathbb{k}[t]$ -module).

PERSISTENT LOWER CENTRAL QUOTIENTS

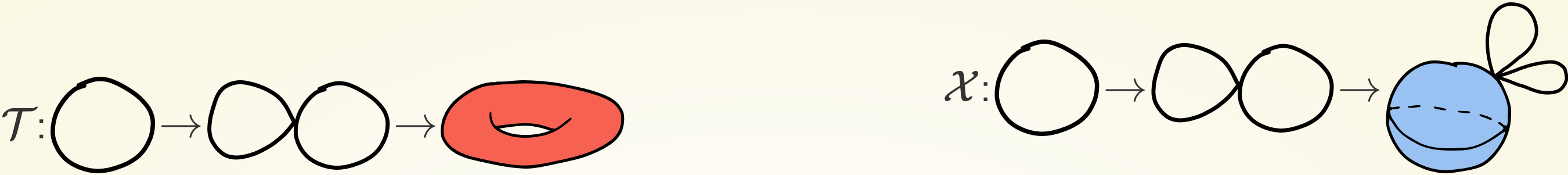
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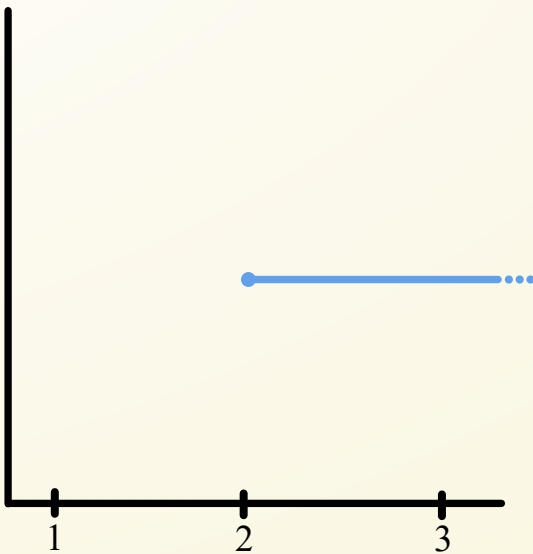
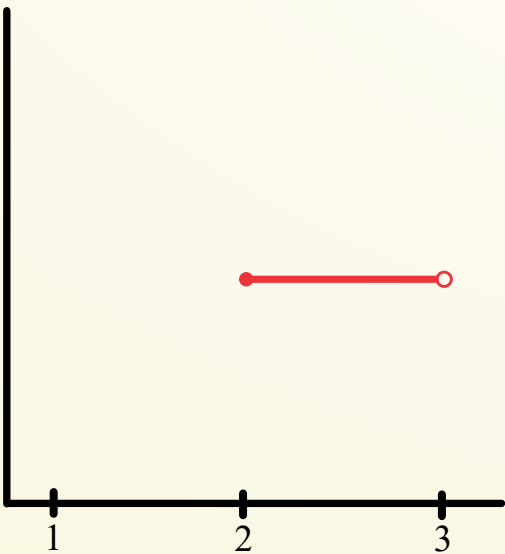
*We can compute **barcodes**. (Skraba & Vejdemo-Johansson, 2013)*

PERSISTENT LOWER CENTRAL QUOTIENTS



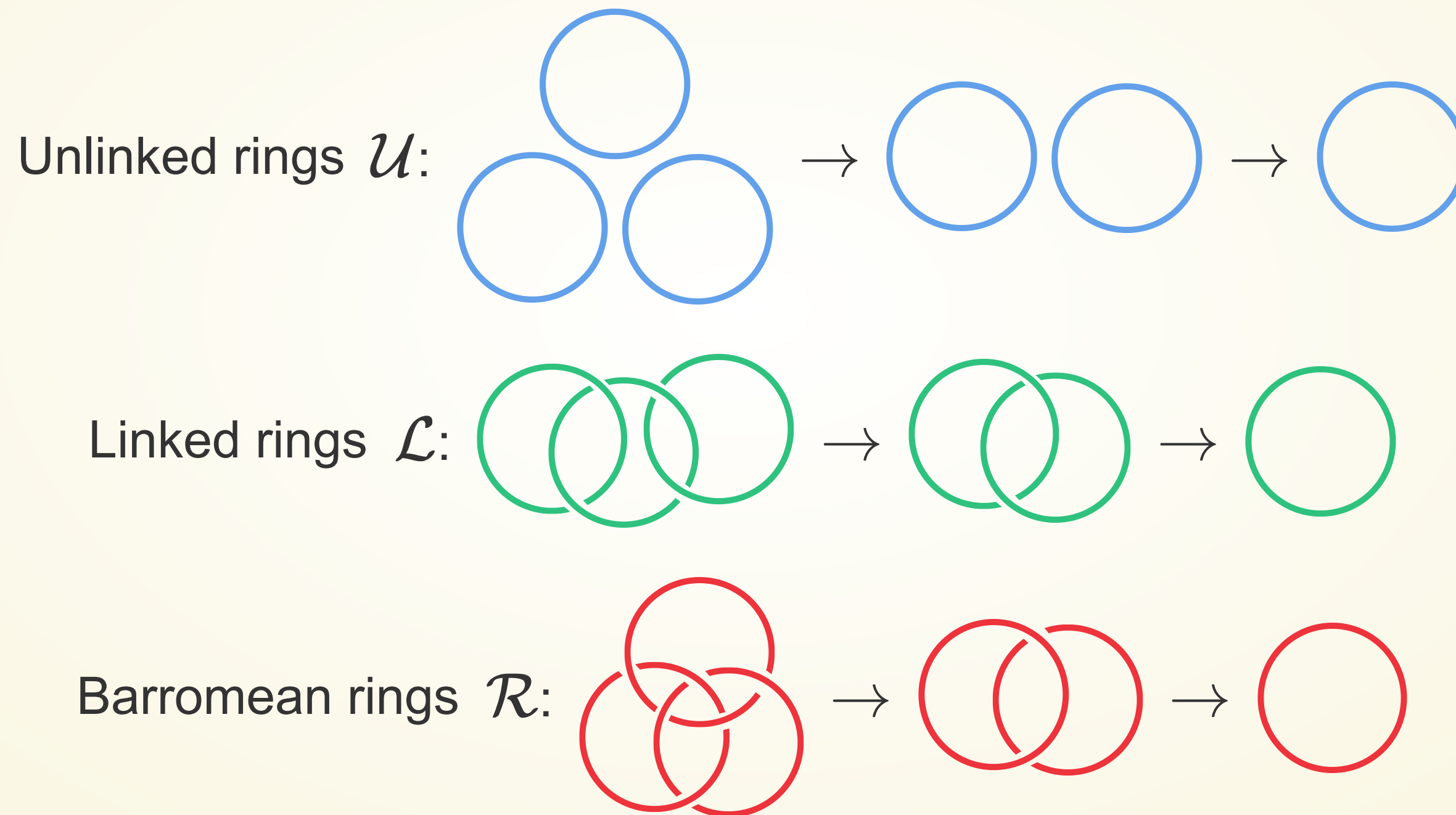
T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3
$\downarrow \pi_1$		$\downarrow \pi_1$		$\downarrow \pi_1$
$\mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} \oplus \mathbb{Z}$
$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$
0	$\longrightarrow$	$\mathbb{k}$	$\longrightarrow$	0

X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow \pi_1$		$\downarrow \pi_1$		$\downarrow \pi_1$
$\mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$
$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$
0	$\longrightarrow$	$\mathbb{k}$	$\longrightarrow$	$\mathbb{k}$

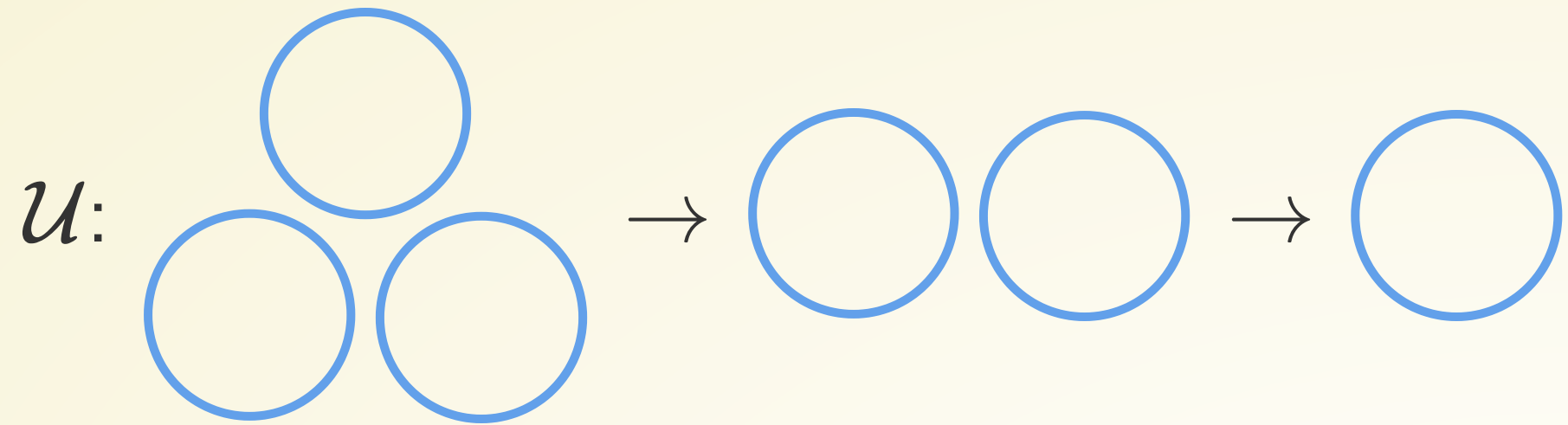


PERSISTENT LOWER CENTRAL QUOTIENTS

Example: lcq_n barcodes of links.



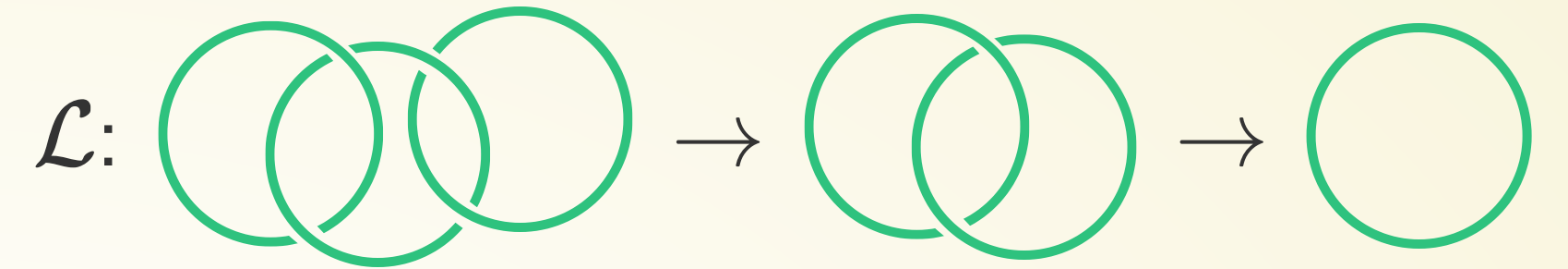
Example: lcq_n barcodes of links. (π_1)



$$\pi_1(\mathbb{R}^3 \setminus U_1) = \langle u_1, u_2, u_3 \mid - \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus U_2) = \langle u_1, u_2 \mid - \rangle$$

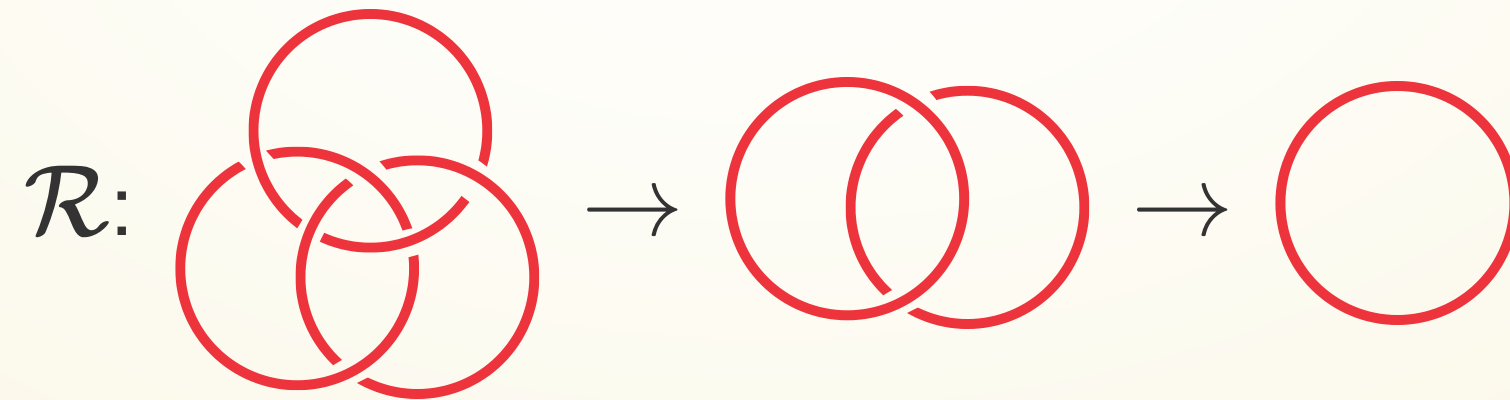
$$\pi_1(\mathbb{R}^3 \setminus U_3) = \langle u_1 \mid - \rangle$$



$$\pi_1(\mathbb{R}^3 \setminus L_1) = \langle l_1, l_2, l_3 \mid [l_1, l_2], [l_1 l_3, l_2] \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus L_2) = \langle l_1, l_2 \mid [l_1, l_2] \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus L_3) = \langle l_1 \mid - \rangle$$

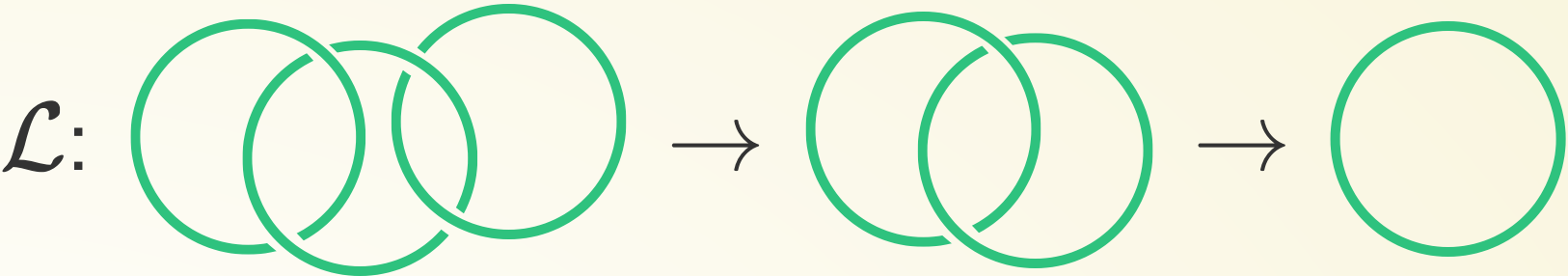
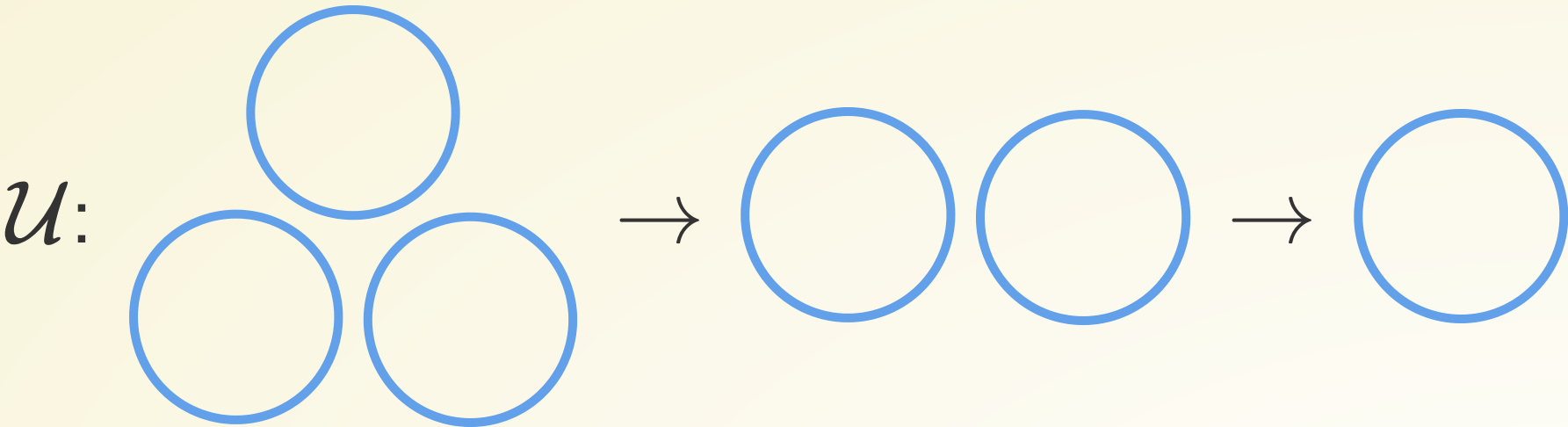


$$\pi_1(\mathbb{R}^3 \setminus R_1) = \langle r_1, r_2, r_3 \mid r_3 r_2^{-1} r_3^{-1} r_1^{-1} r_3 r_2 r_3^{-1} r_2^{-1} r_1 r_2, r_3 r_2 r_3^{-1} r_1 r_3 r_1^{-1} r_2^{-1} r_1 r_3^{-1} r_1^{-1} \rangle$$

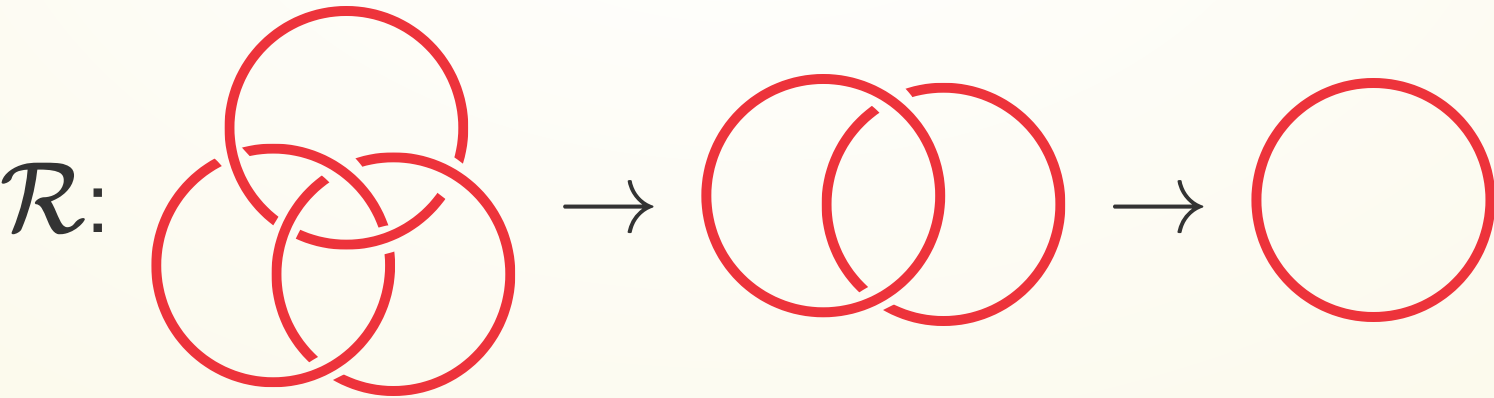
$$\pi_1(\mathbb{R}^3 \setminus R_2) = \langle r_1, r_2 \mid - \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus R_3) = \langle r_1 \mid - \rangle$$

Example: lcq_n barcodes of links. $(lcq_1^{\mathbb{k}})$

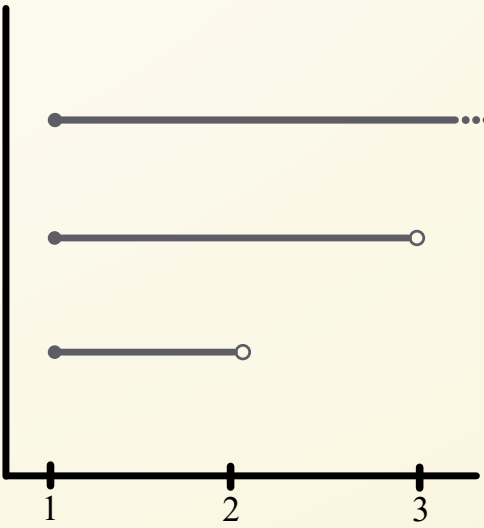
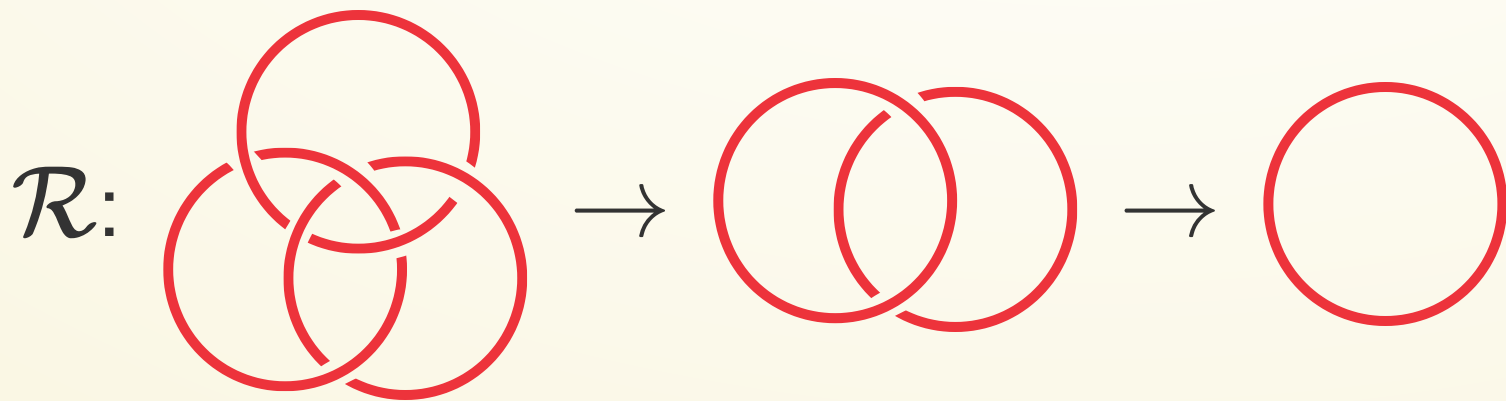
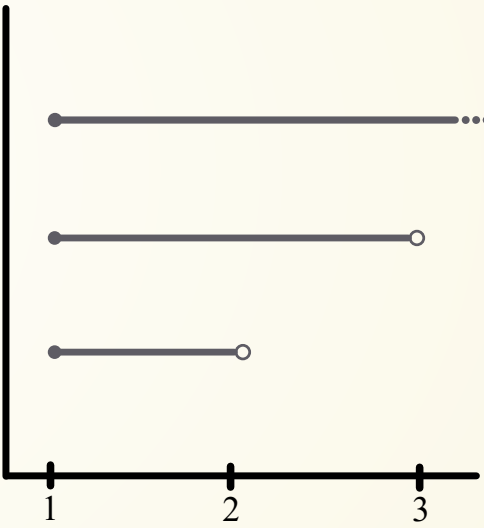
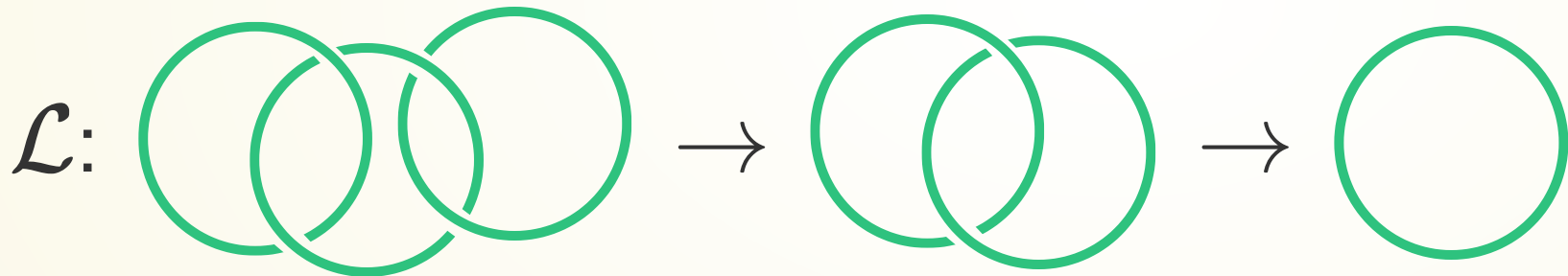
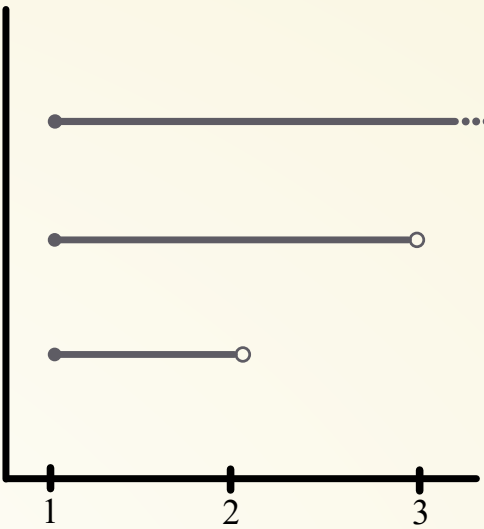
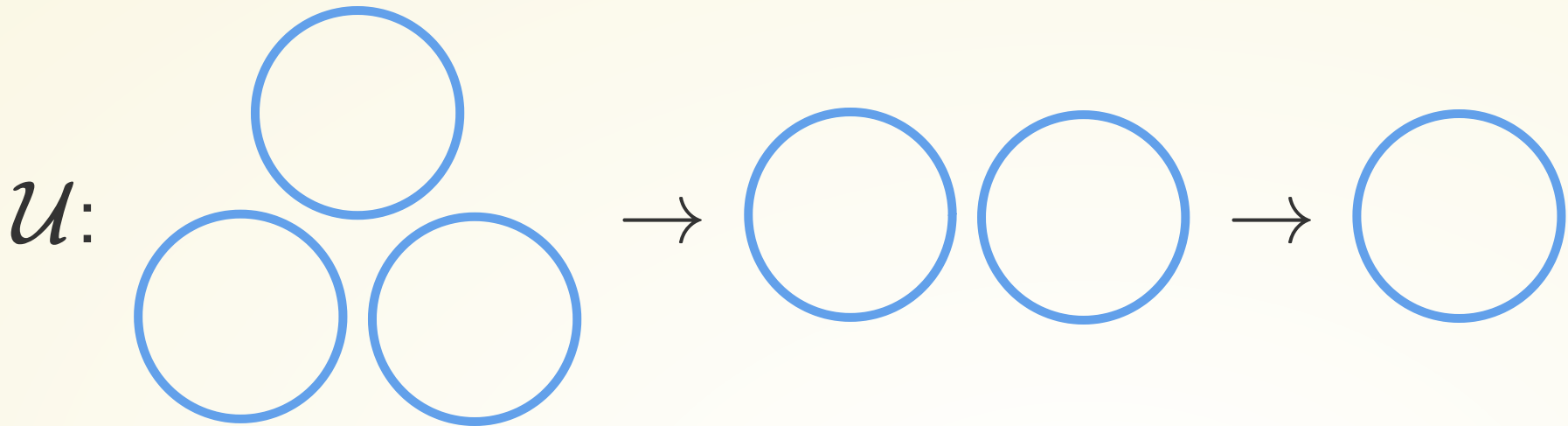


$\mathbb{R}^3 \setminus U_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_3$	$\mathbb{R}^3 \setminus L_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_3$
$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$	$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$
$\mathbb{k}^3$	$\longrightarrow$	$\mathbb{k}^2$	$\longrightarrow$	$\mathbb{k}$	$\mathbb{k}^3$	$\longrightarrow$	$\mathbb{k}^2$	$\longrightarrow$	$\mathbb{k}$

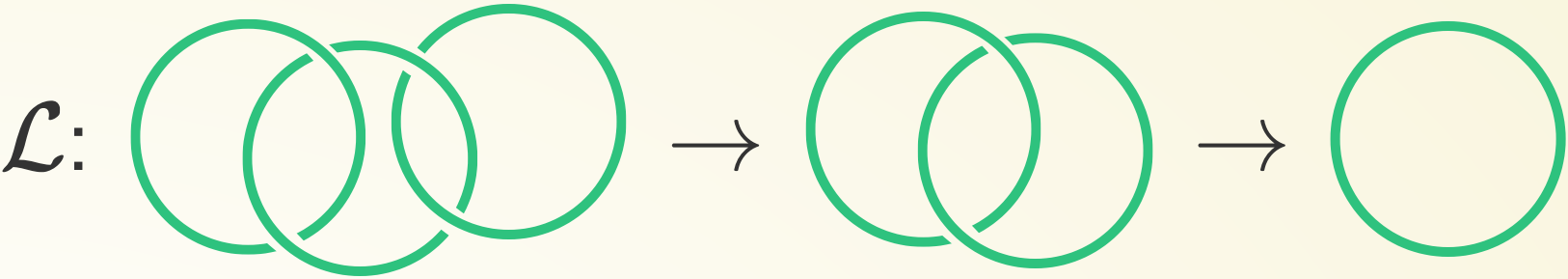
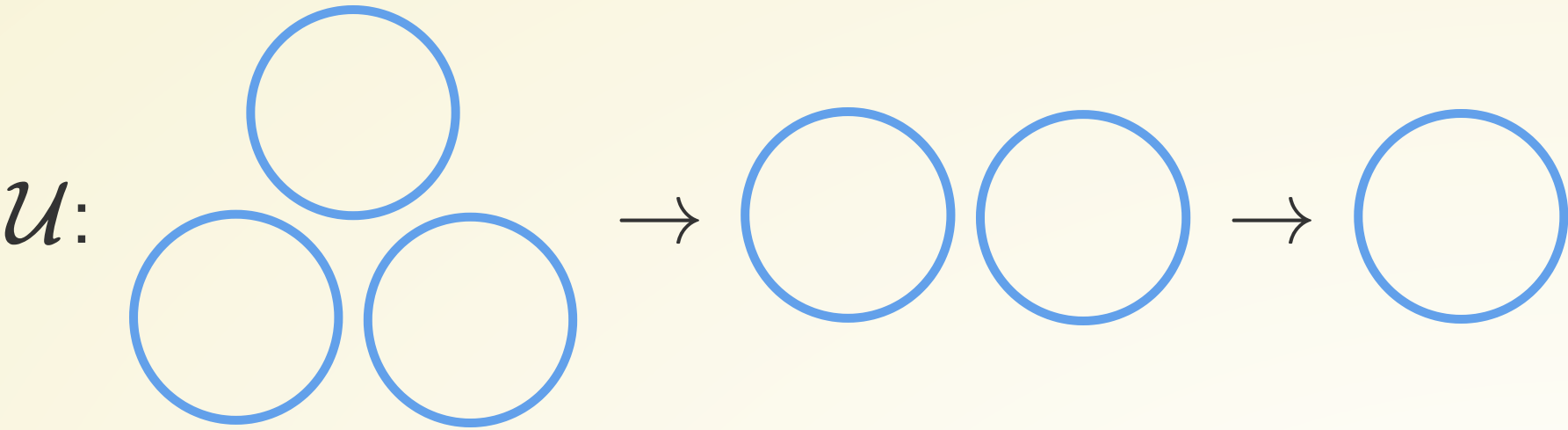


$\mathbb{R}^3 \setminus R_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_3$
$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$
$\mathbb{k}^3$	$\longrightarrow$	$\mathbb{k}^2$	$\longrightarrow$	$\mathbb{k}$

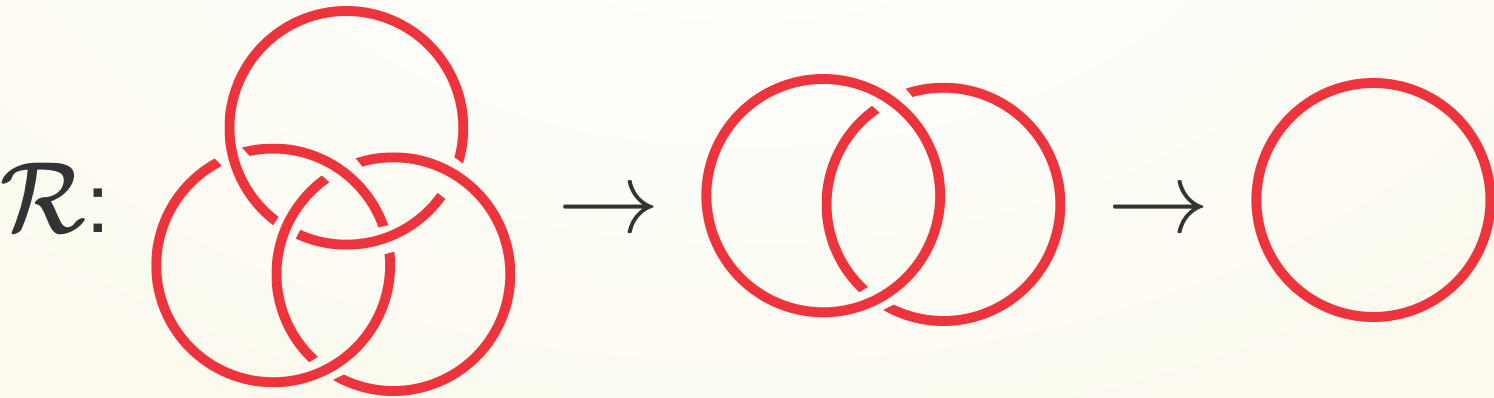
Example: lcq_n barcodes of links. $(lcq_1^{\mathbb{K}})$



Example: lcq_n barcodes of links. $(lcq_2^{\mathbb{k}})$

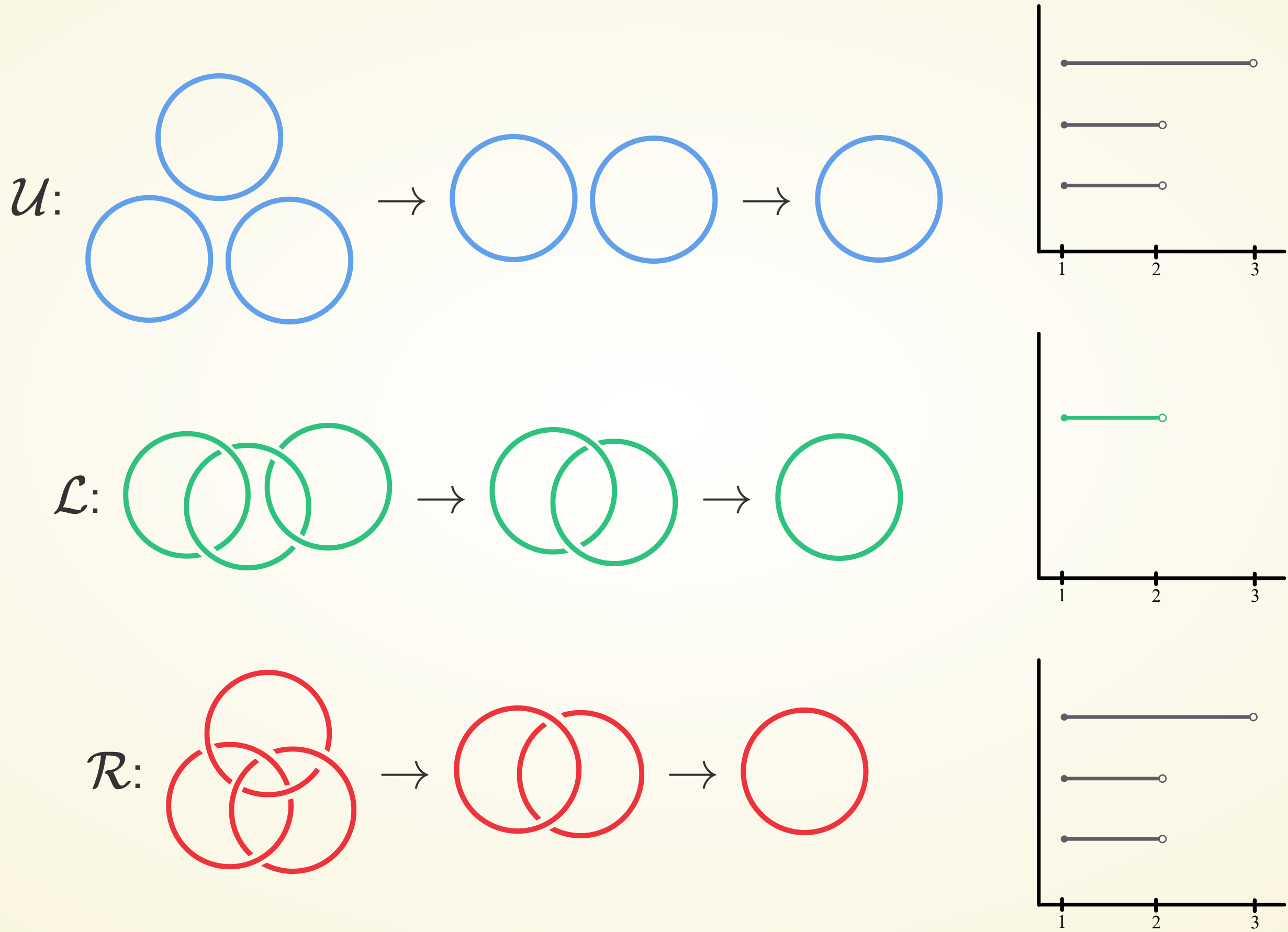


$\mathbb{R}^3 \setminus U_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_3$	$\mathbb{R}^3 \setminus L_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_3$
$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$	$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$
$\mathbb{k}^3$	$\longrightarrow$	$\mathbb{k}$	$\longrightarrow$	0	$\mathbb{k}$	$\longrightarrow$	0	$\longrightarrow$	0

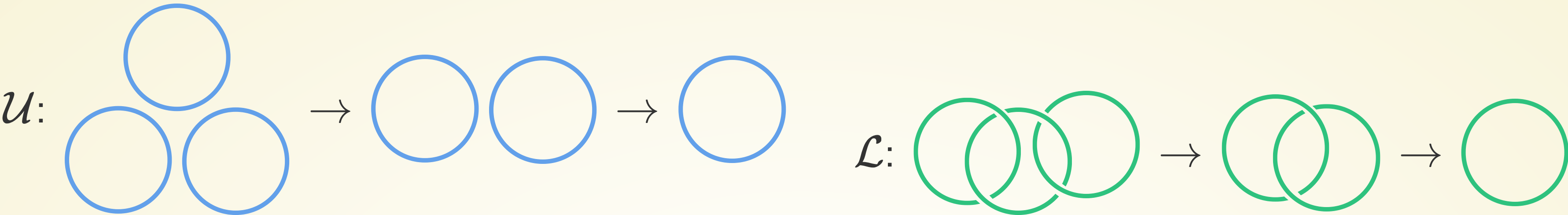


$\mathbb{R}^3 \setminus R_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_3$
$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$		$\downarrow lcq_2^{\mathbb{k}}$
$\mathbb{k}^3$	$\longrightarrow$	$\mathbb{k}$	$\longrightarrow$	0

Example: lcq_n barcodes of links. $(lcq_2^{\mathbb{K}})$



Example: lcq_n barcodes of links. $(lcq_3^{\mathbb{k}})$



$\mathbb{R}^3 \setminus U_1$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus U_2$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus U_3$

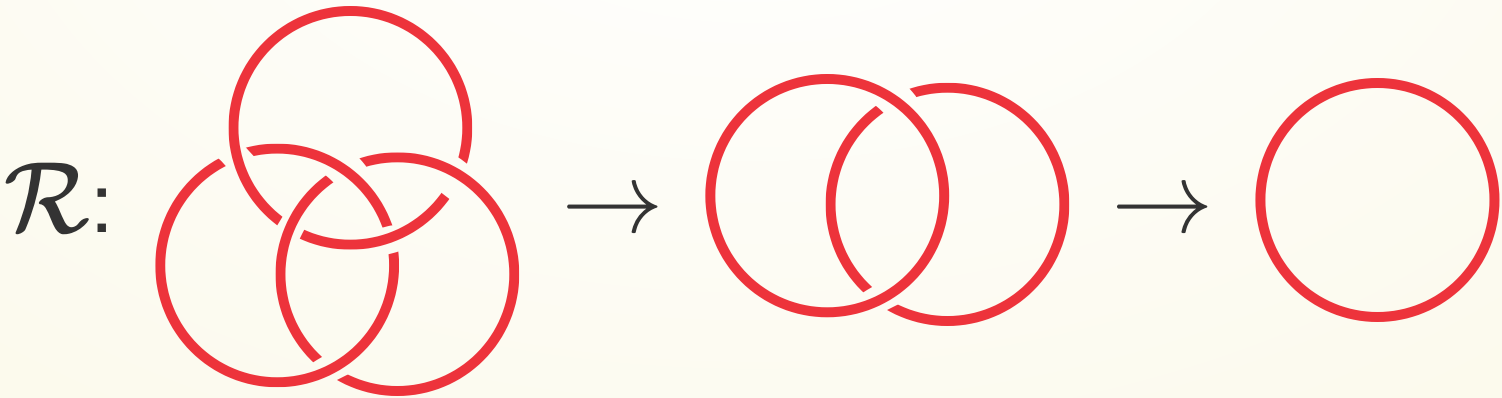
$\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$

$\mathbb{k}^8$
 $\longrightarrow$
 $\mathbb{k}^2$
 $\longrightarrow$
 0

$\mathbb{R}^3 \setminus L_1$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus L_2$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus L_3$

$\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$

$\mathbb{k}^2$
 $\longrightarrow$
 0
 $\longrightarrow$
 0

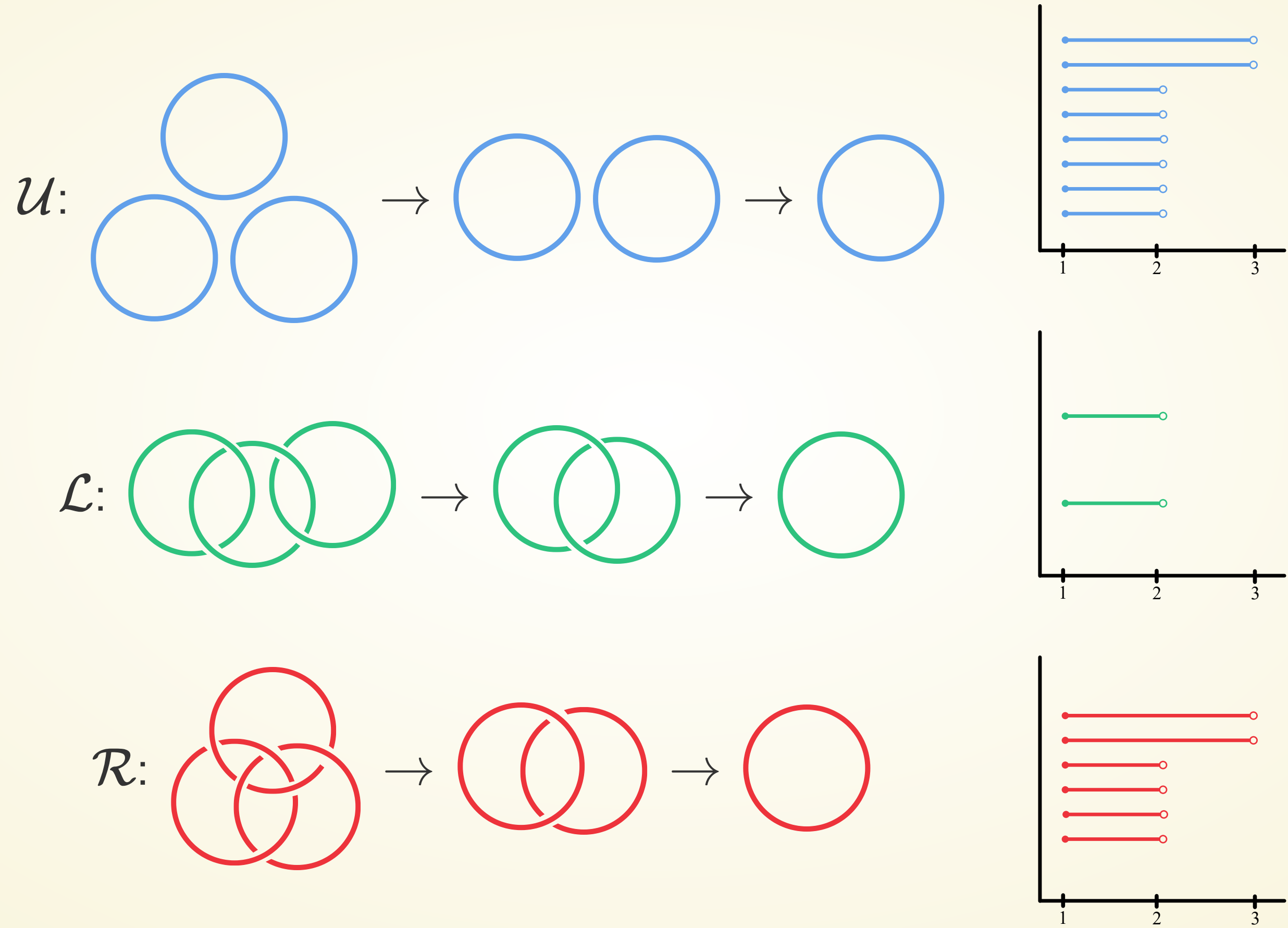


$\mathbb{R}^3 \setminus R_1$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus R_2$
 $\longrightarrow$
 $\mathbb{R}^3 \setminus R_3$

$\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$
 $\downarrow lcq_3^{\mathbb{k}}$

$\mathbb{k}^6$
 $\longrightarrow$
 $\mathbb{k}^2$
 $\longrightarrow$
 0

Example: lcq_n barcodes of links. $(lcq_3^{\mathbb{K}})$



GAP CODE

```
1 LoadPackage("nq");
2
3 LcsQuotient := function(group, i)
4   local nq, lcs;
5   nq := NilpotentQuotient(group, i+1);
6   lcs := LowerCentralSeries(nq);
7   if i < Length(lcs) then
8     return AbelianInvariants(lcs[i] / lcs[i+1]);
9   fi;
10  return TrivialGroup();
11 end;
```

Using GAP 4 NQ Package (Nickel, 2007).

SUMMARY

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- We constructed **lower central quotient barcodes** for persistent π_1 .
- This invariant retains **non-commutative information** from persistent π_1 .

$$\pi_1 \rightsquigarrow lcq_n \rightsquigarrow lcq_n^{\mathbb{K}} \rightsquigarrow \text{Barcodes}$$

ONGOING AND FUTURE WORK

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